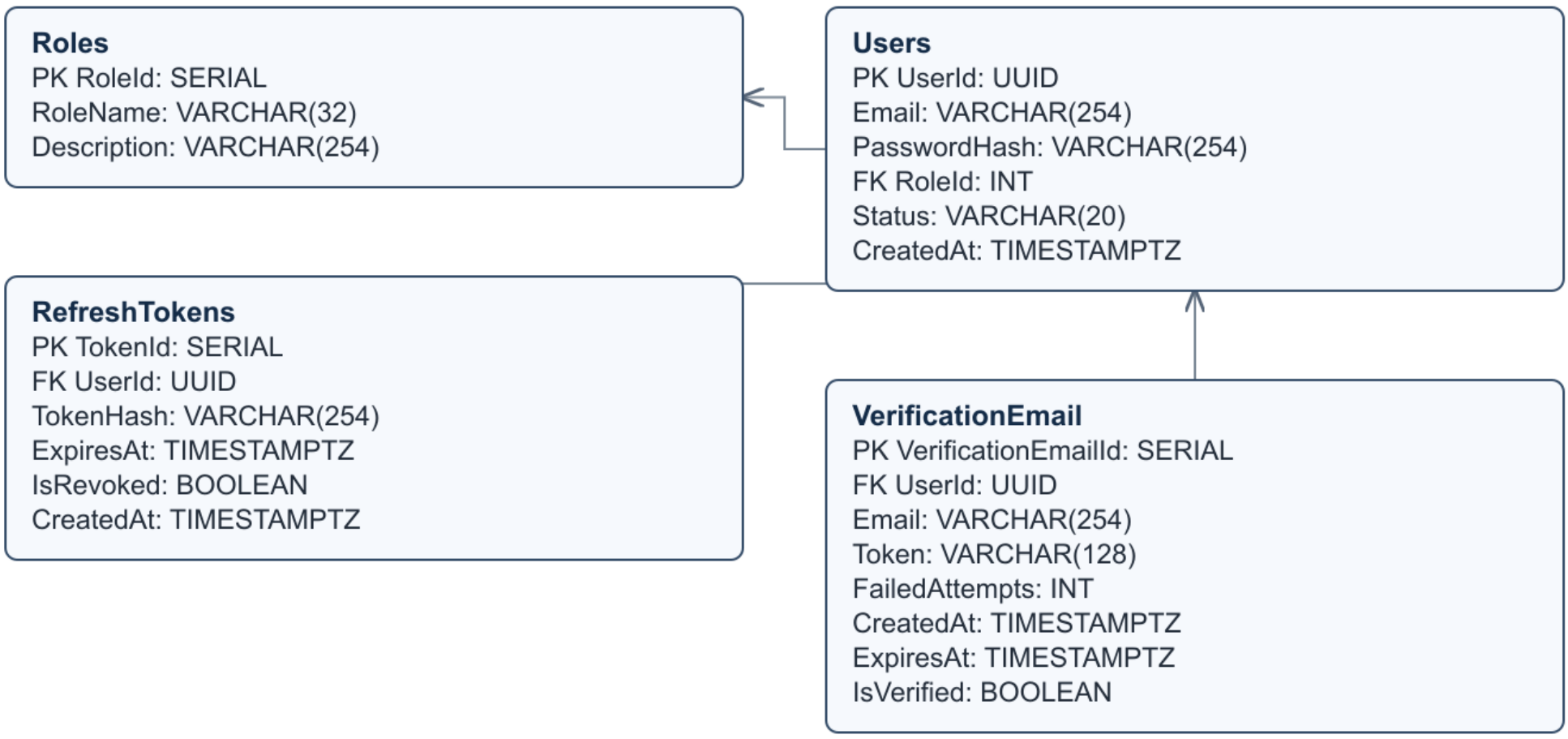
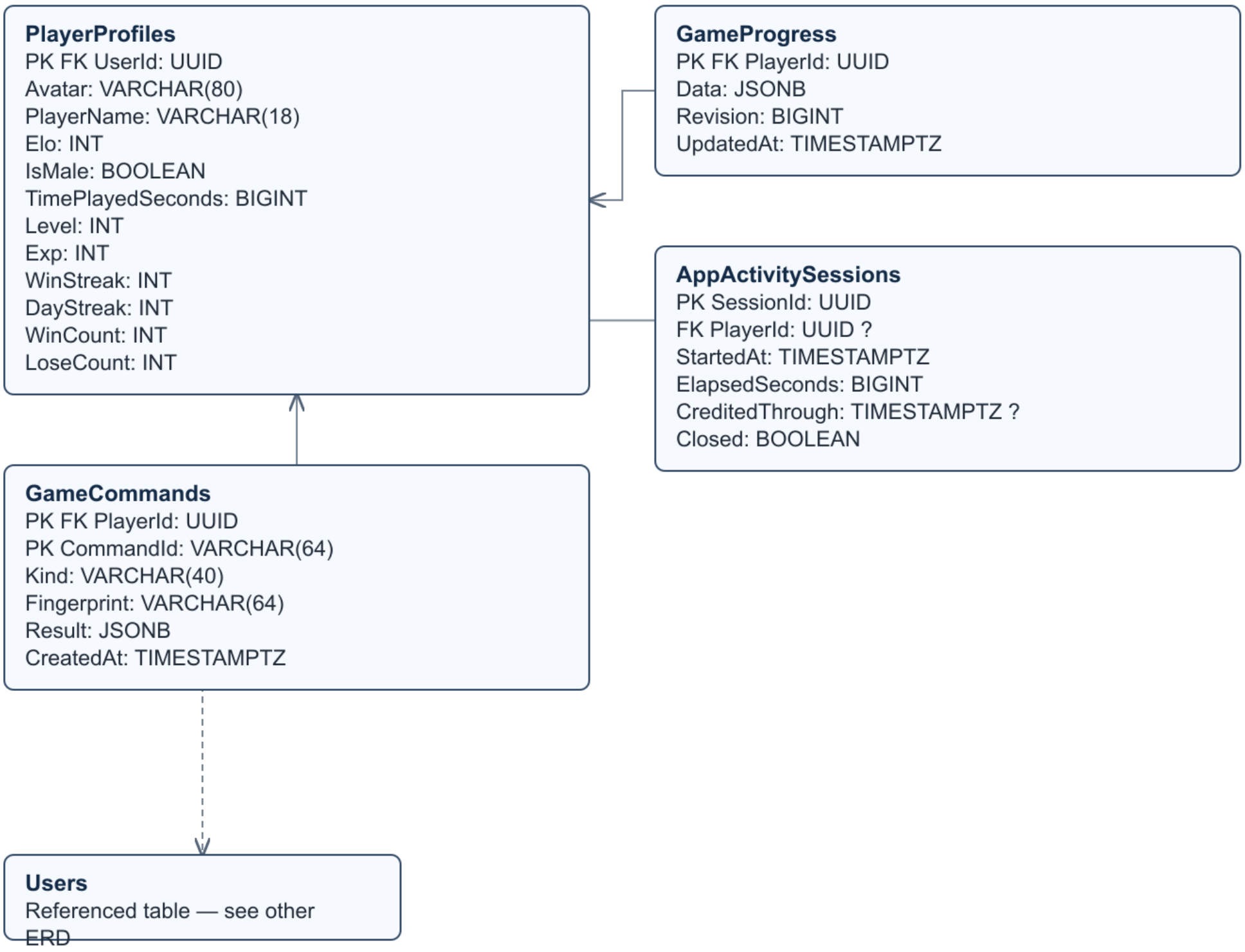
ERD01 Current SQL schema 1

37 tables / 208 columns / 54 declared foreign keys. Dashed link: table shown on another page.



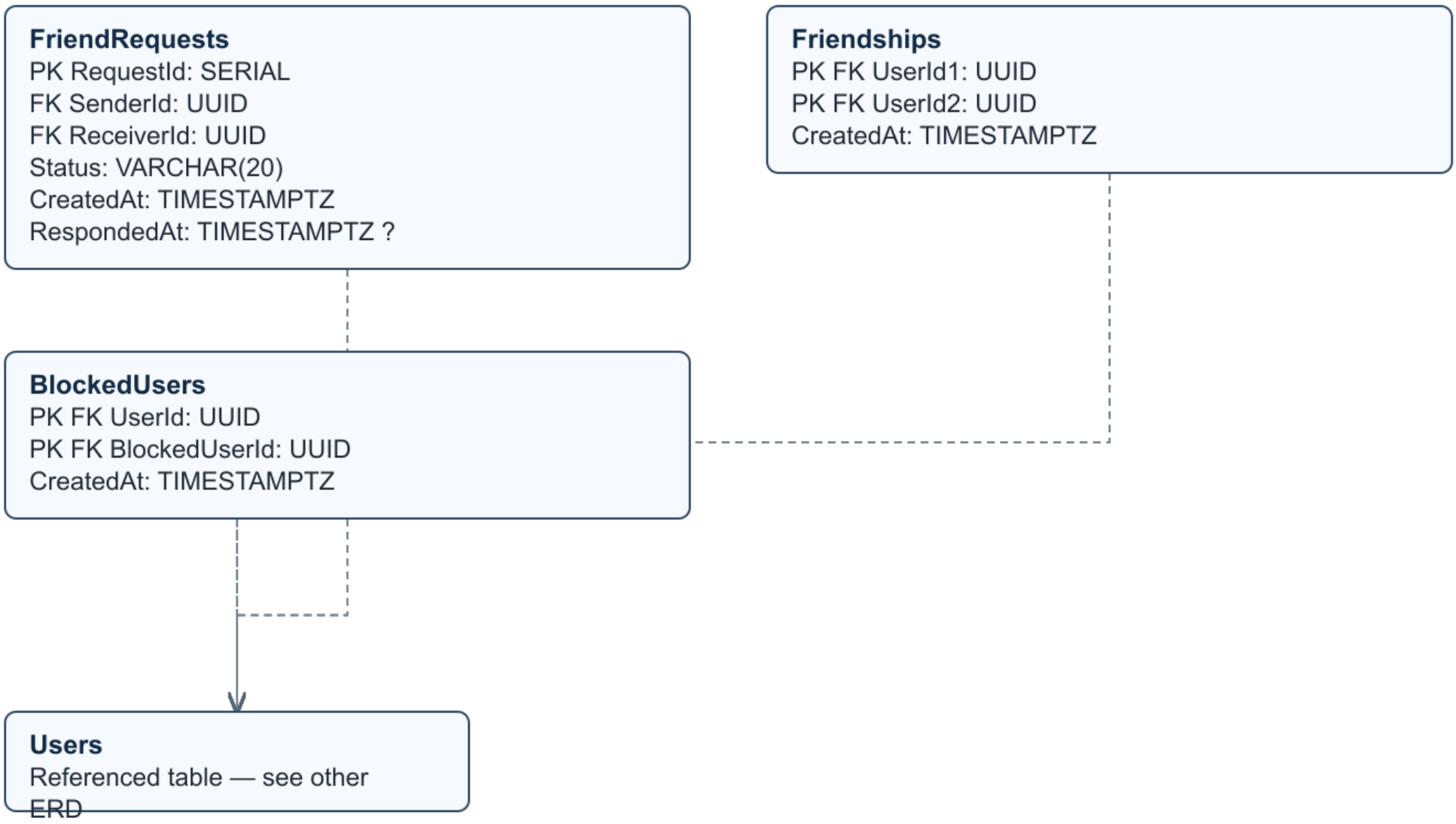
ERD02 Current SQL schema 2

37 tables / 208 columns / 54 declared foreign keys. Dashed link: table shown on another page.



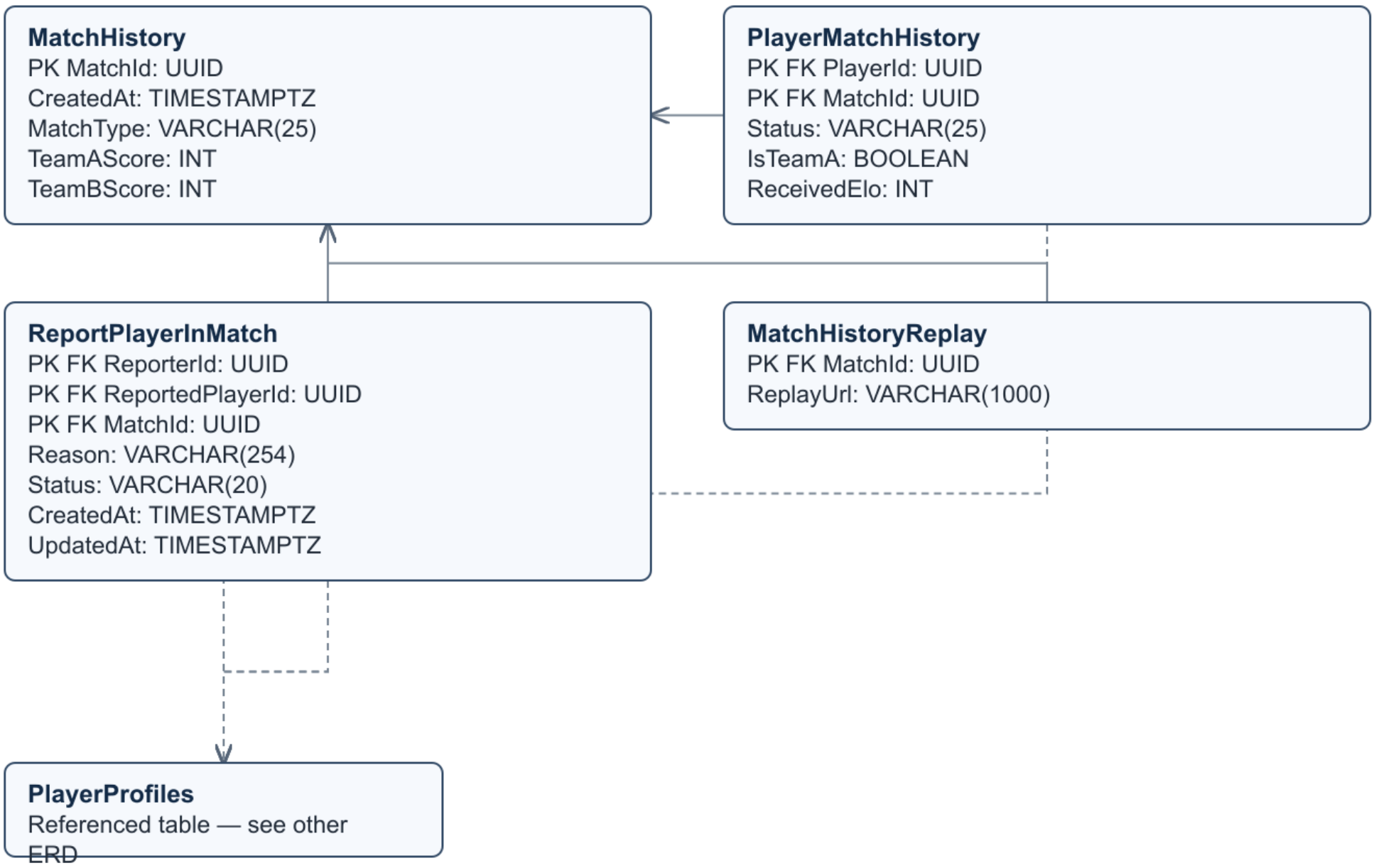
ERD03 Current SQL schema 3

37 tables / 208 columns / 54 declared foreign keys. Dashed link: table shown on another page.



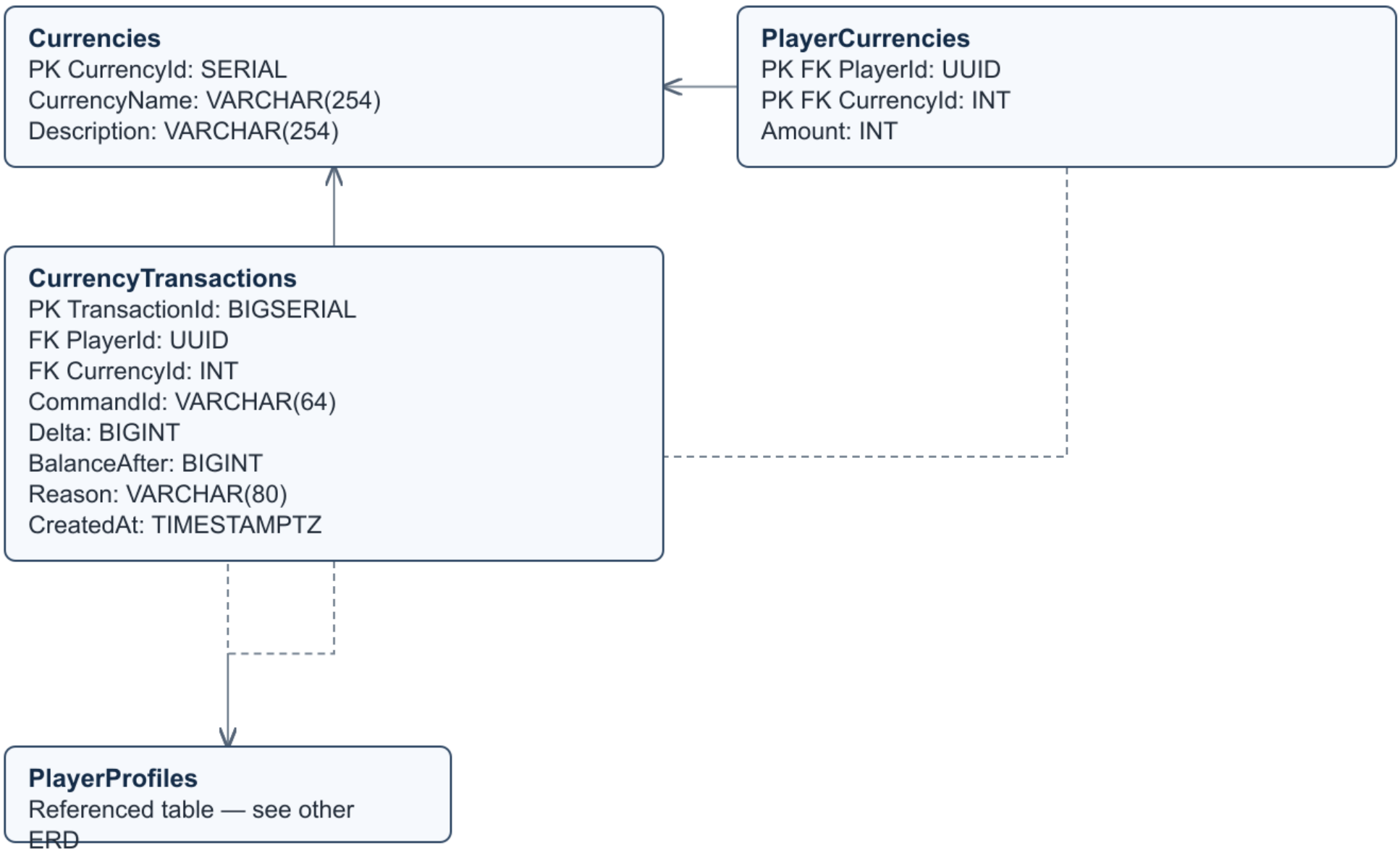
ERD04 Current SQL schema 4

37 tables / 208 columns / 54 declared foreign keys. Dashed link: table shown on another page.



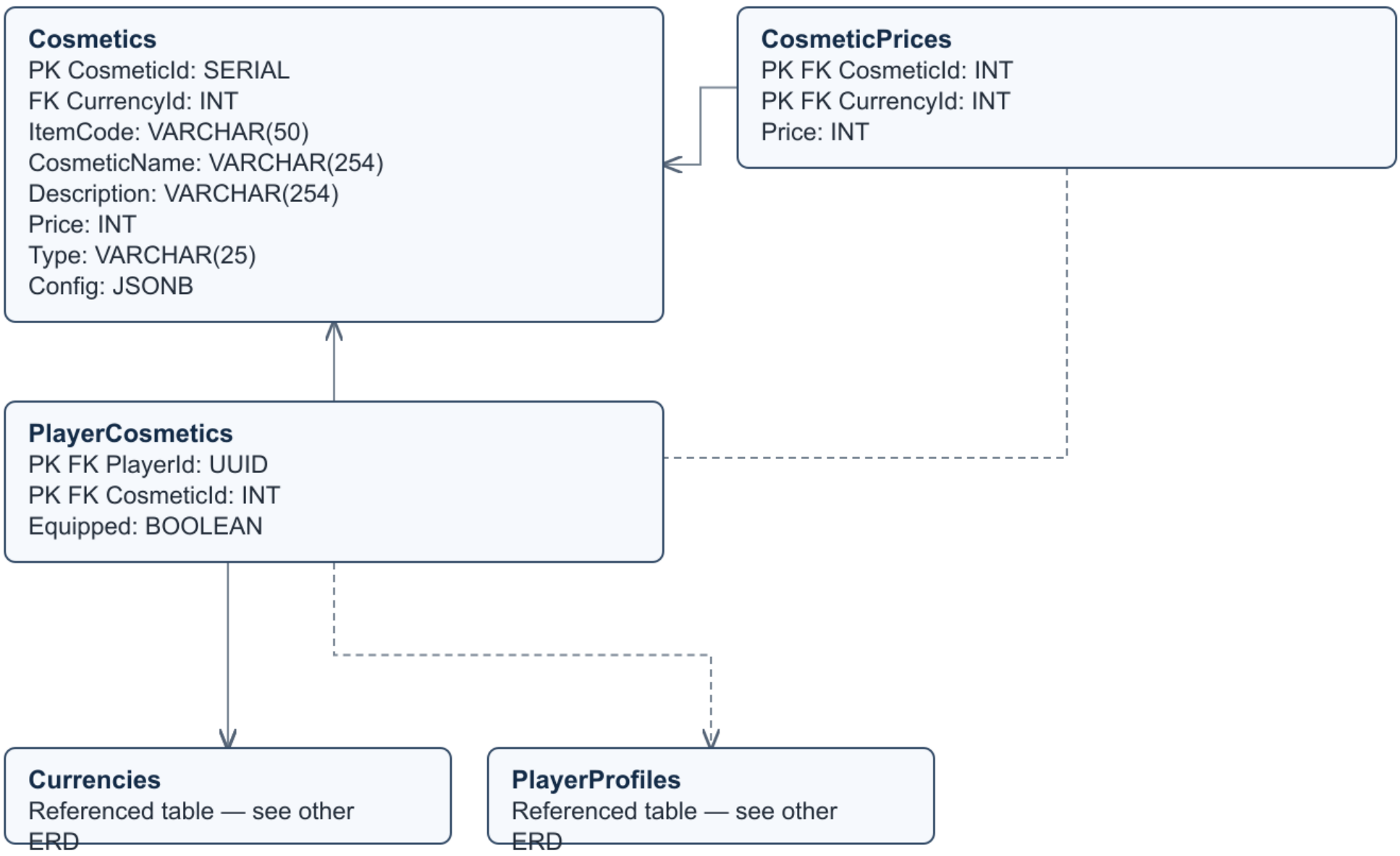
ERD05 Current SQL schema 5

37 tables / 208 columns / 54 declared foreign keys. Dashed link: table shown on another page.



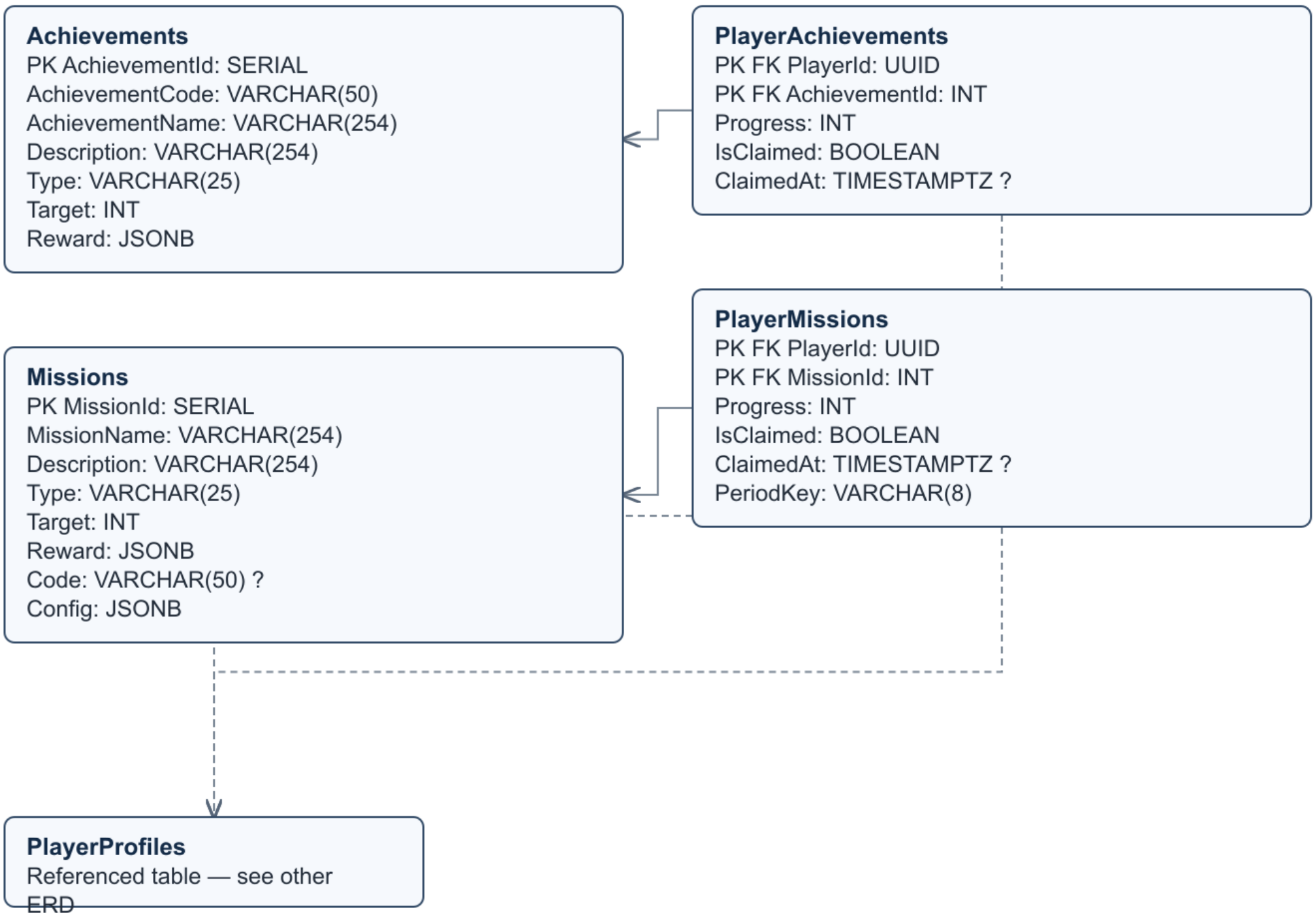
ERD06 Current SQL schema 6

37 tables / 208 columns / 54 declared foreign keys. Dashed link: table shown on another page.



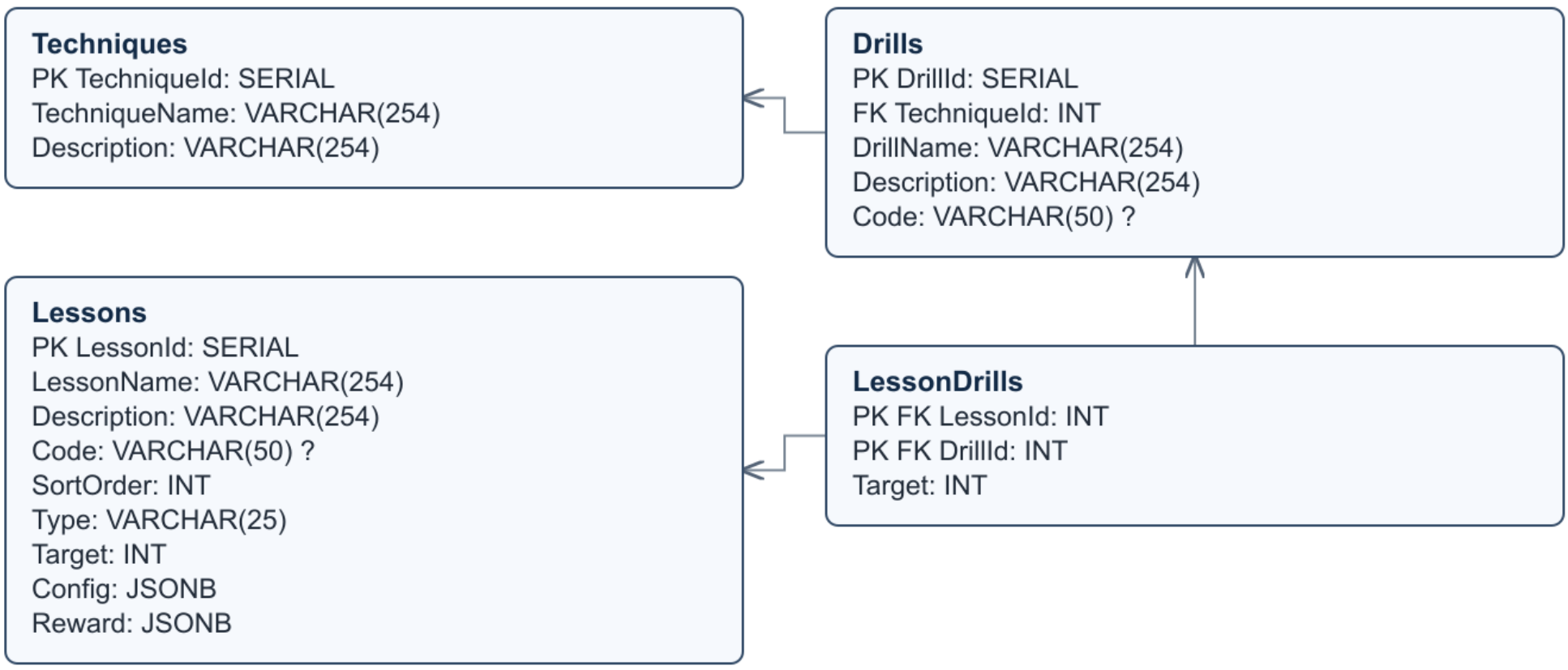
ERD07 Current SQL schema 7

37 tables / 208 columns / 54 declared foreign keys. Dashed link: table shown on another page.



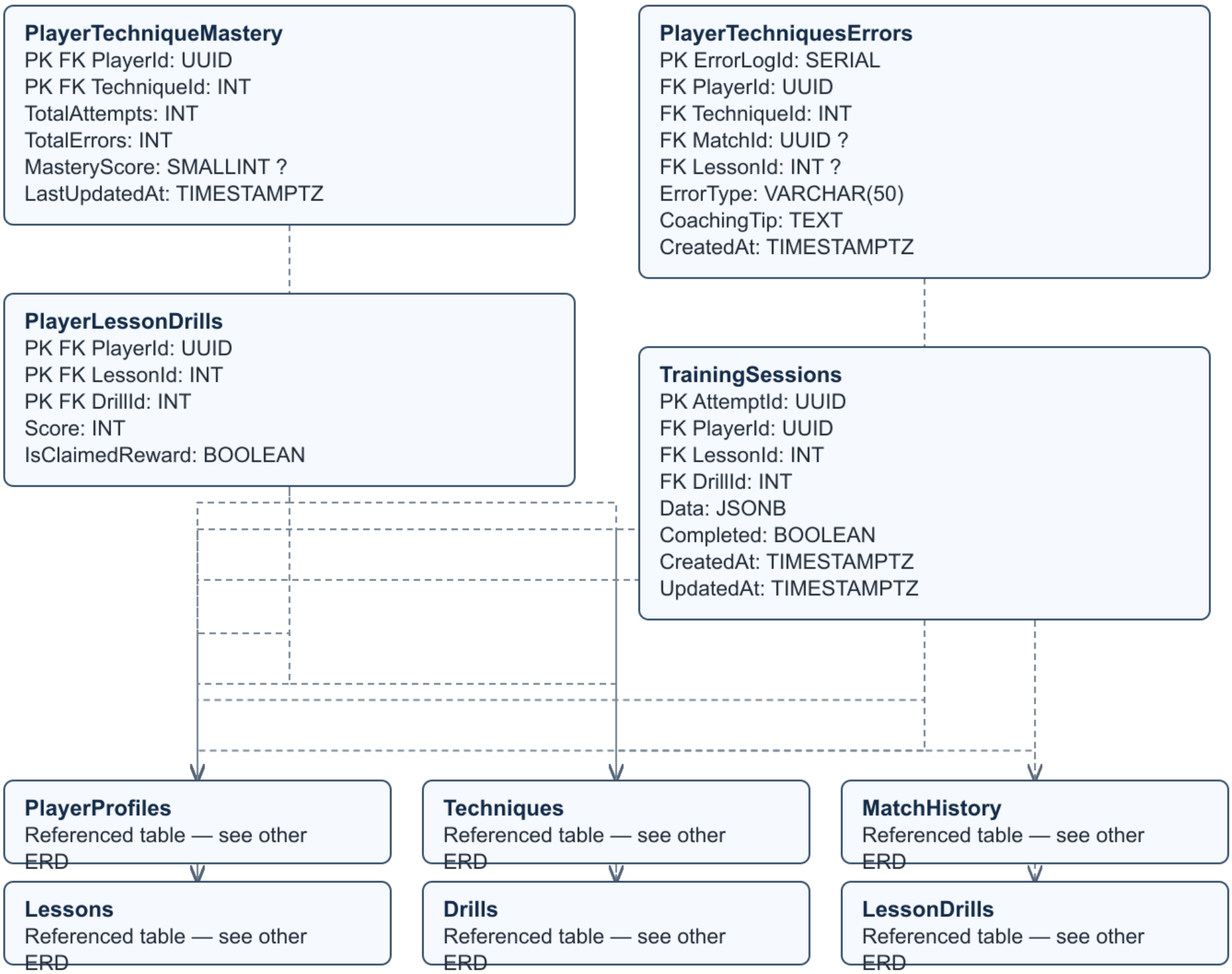
ERD08 Current SQL schema 8

37 tables / 208 columns / 54 declared foreign keys. Dashed link: table shown on another page.



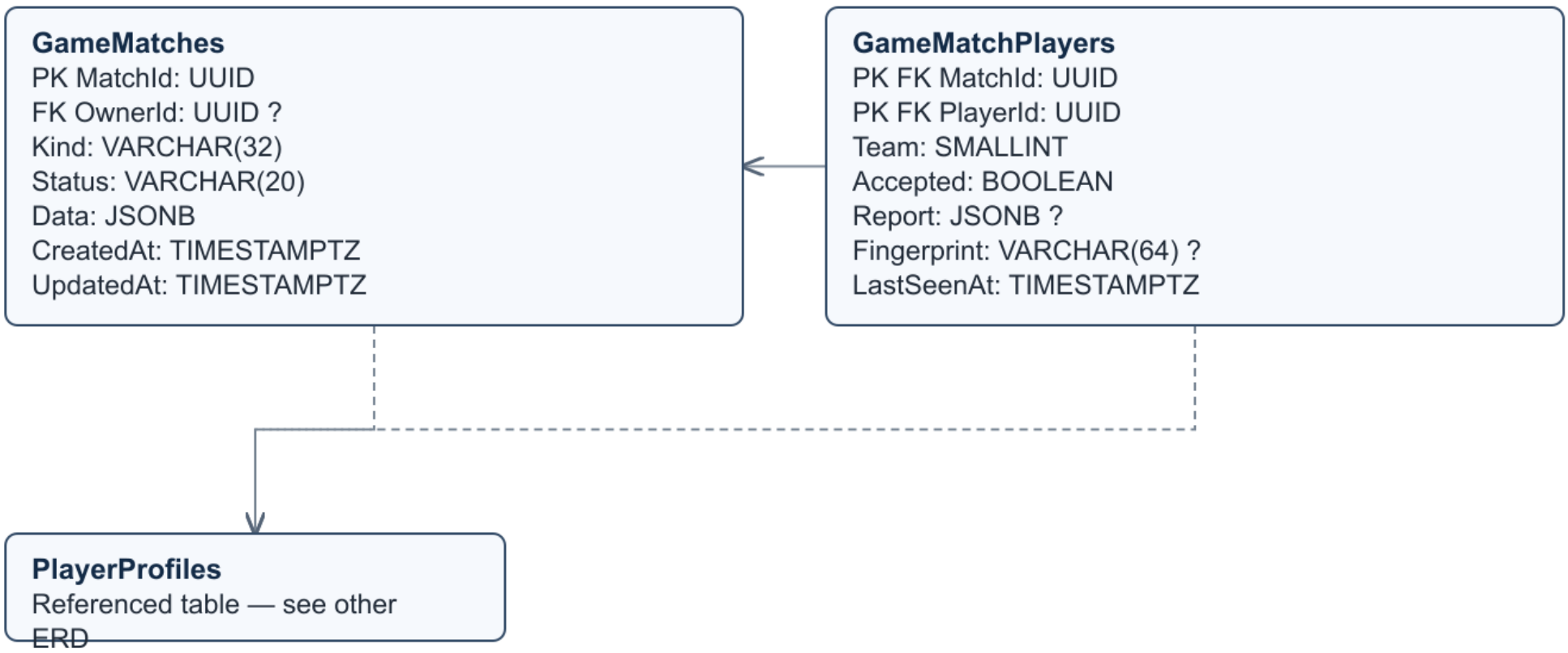
ERD09 Current SQL schema 9

37 tables / 208 columns / 54 declared foreign keys. Dashed link: table shown on another page.



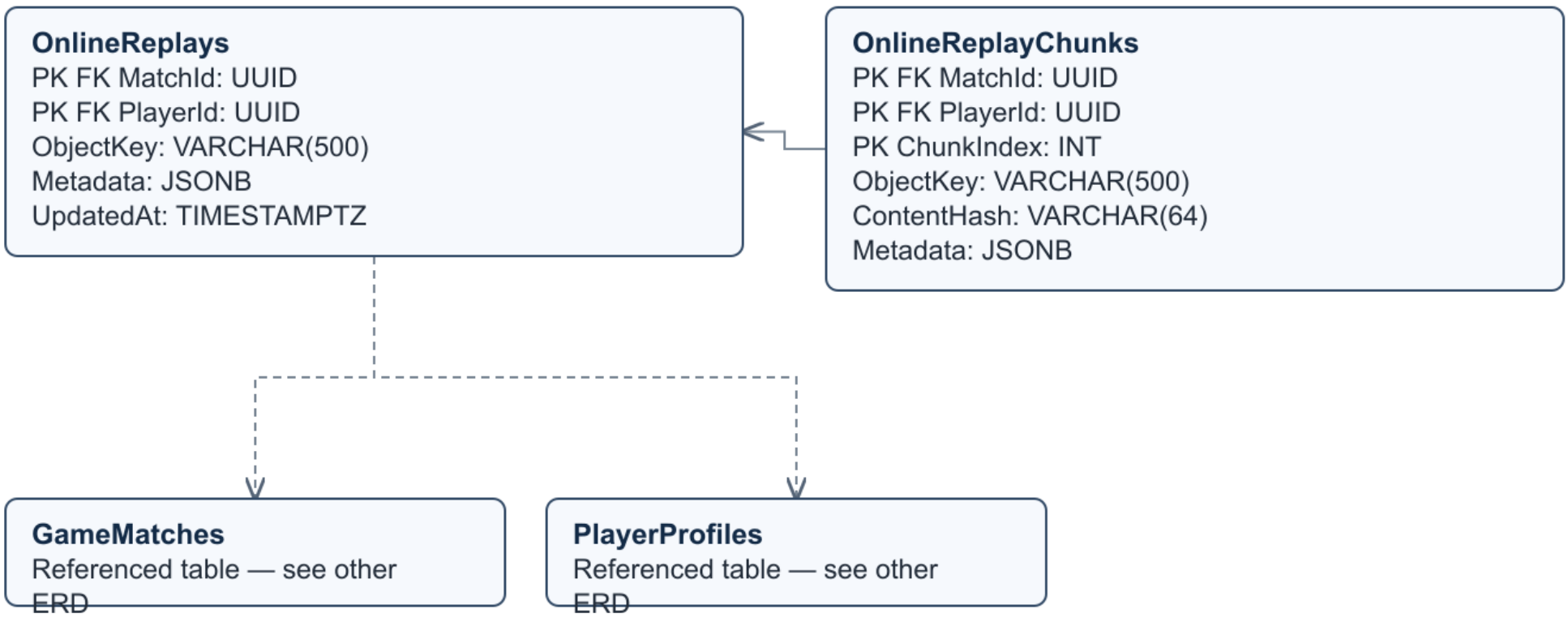
ERD10 Current SQL schema 10

37 tables / 208 columns / 54 declared foreign keys. Dashed link: table shown on another page.



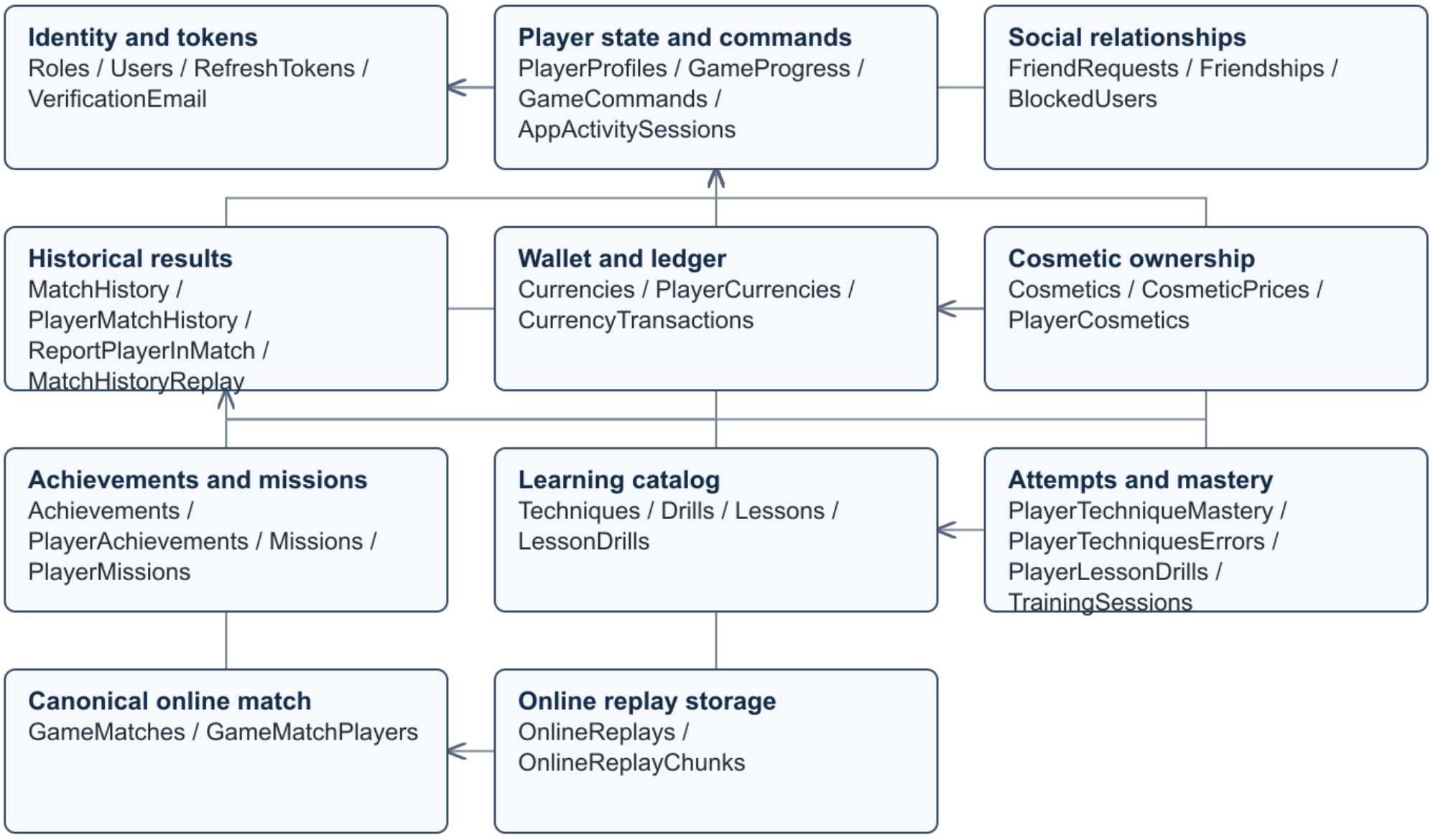
ERD11 Current SQL schema 11

37 tables / 208 columns / 54 declared foreign keys. Dashed link: table shown on another page.



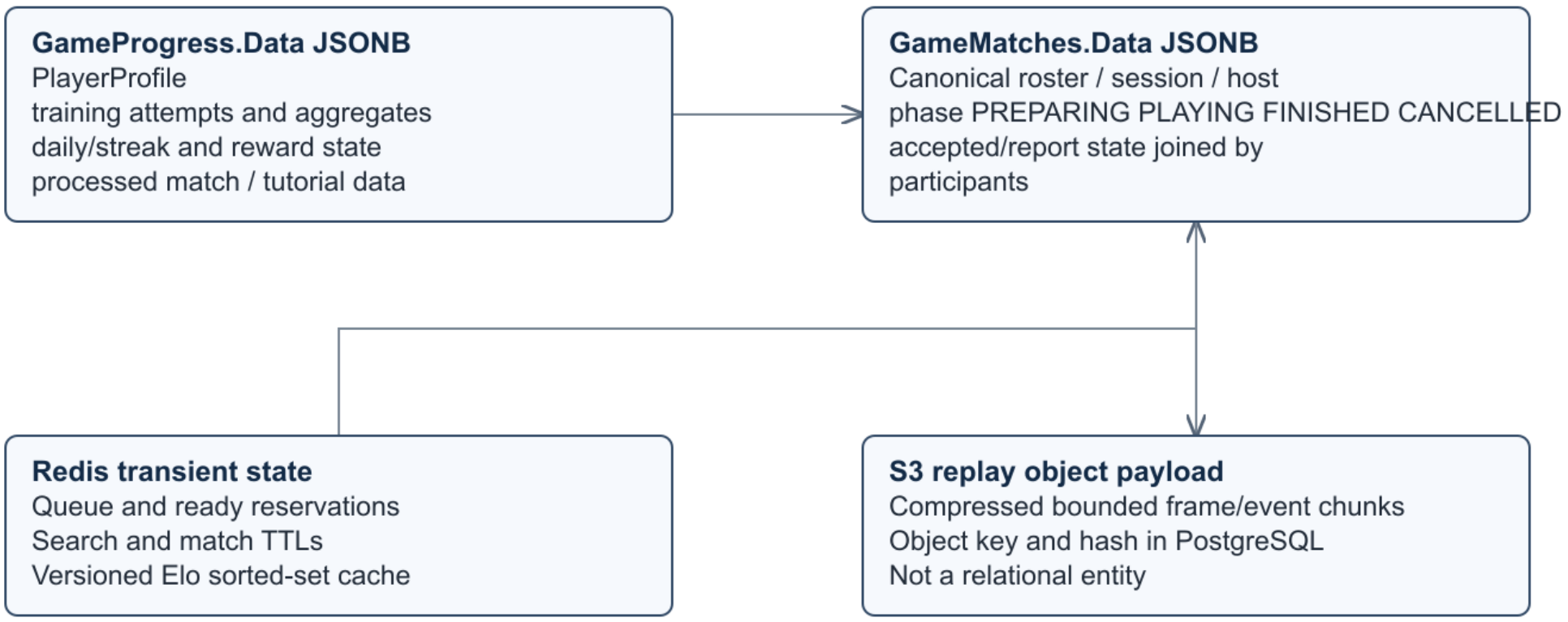
ERD00 Relational data domains at the pulled revision

Arrows point from referencing to referenced domain. This overview omits attributes; ERD01–11 and the CSV are complete.



DATA01 JSON and transient data ownership

JSON members are schema-less persisted payloads, not extra SQL tables. PostgreSQL owns durable state; Redis TTL loss must not award rewards.



LOGICAL01 Logical entities and association meaning

Conceptual cardinalities are explained inside each domain. ERD01–11 map each association to explicit SQL keys; optionality and delete policy are in the FK register.

Account and player
User 1 — 0..1 PlayerProfile row
Role 1 — many Users
Profile 1 — many owned records

Learning catalog and attempt
Lesson many — many Drill via LessonDrills
Technique 1 — many Drills
Player 1 — many TrainingSessions

Match and participant result
GameMatch 1 — many GameMatchPlayers
History 1 — many PlayerMatchHistory
Player many — many Match via participants

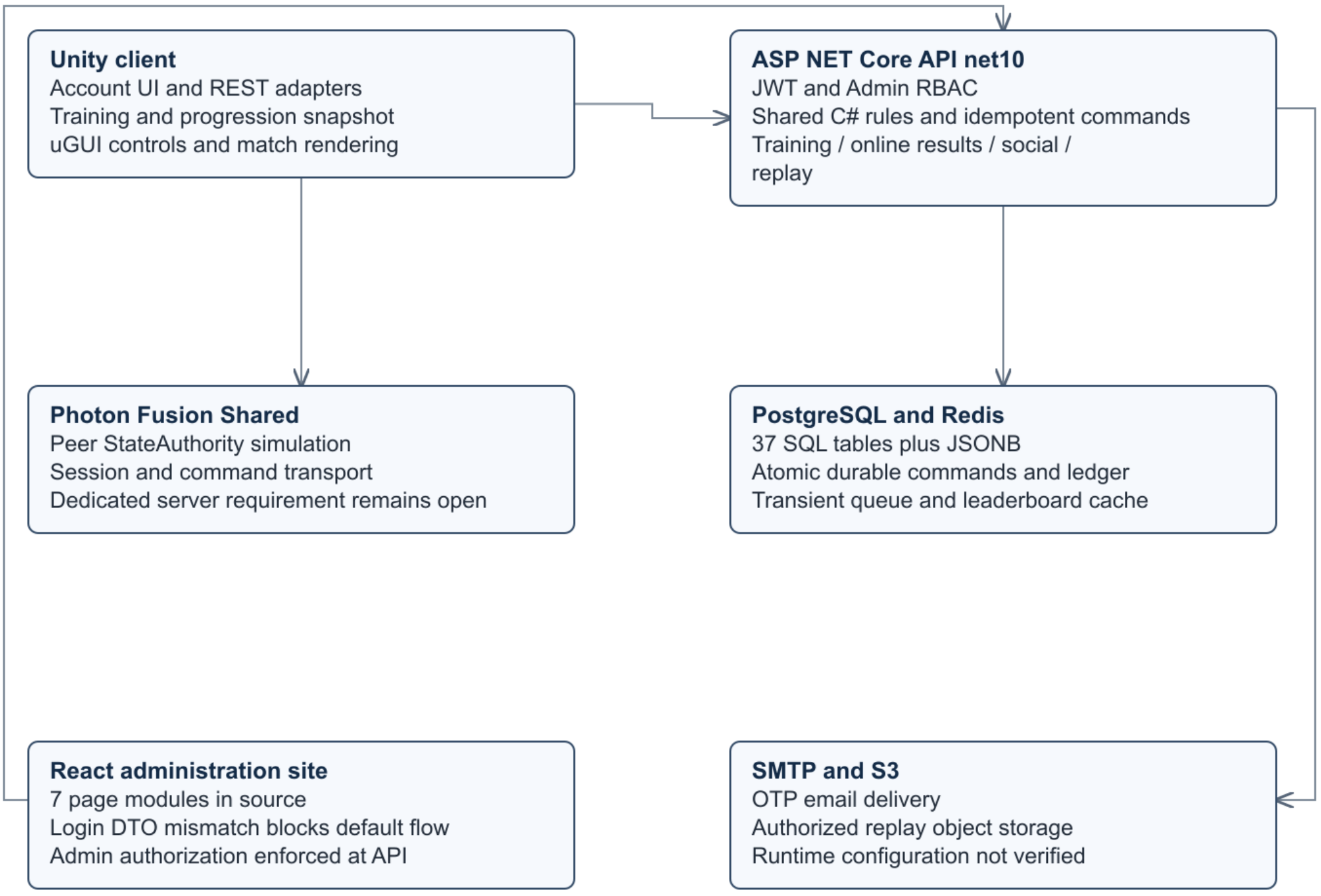
Economy and objectives
Player many — many Cosmetic via ownership
Player many — many Mission/Achievement
Wallet 1 currency/player; ledger many entries

Social relationship
Users many — many via Friendship
Request has exactly one sender and recipient
Block has owner and blocked user

Replay and progress aggregate
Match/player 1 — many replay chunks
Player 1 — 0..1 GameProgress row
JSONB aggregate members are not SQL tables

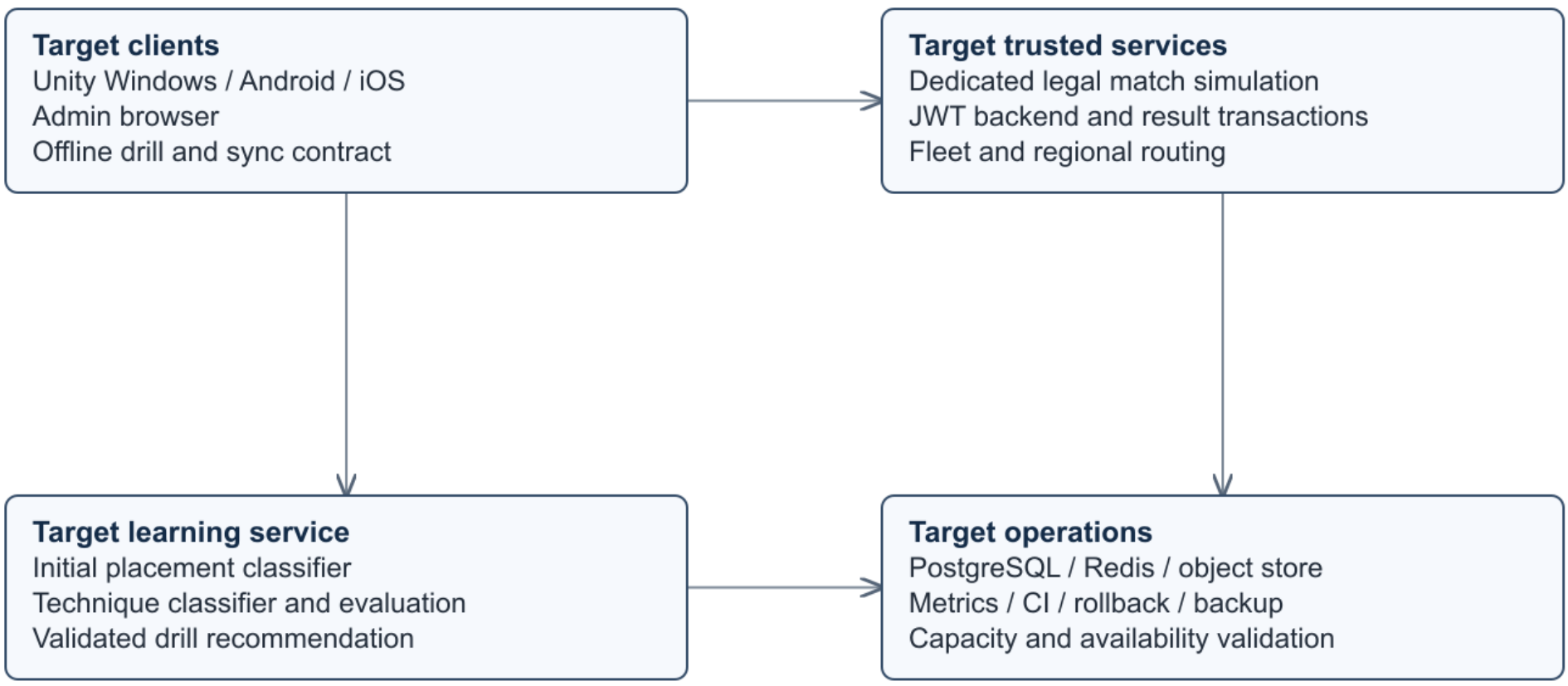
ARCH01 Architecture visible in current source

Current source is not proof of a deployed environment. Old UGS/NGO adapters remaining in the repository are not the active Fusion path.



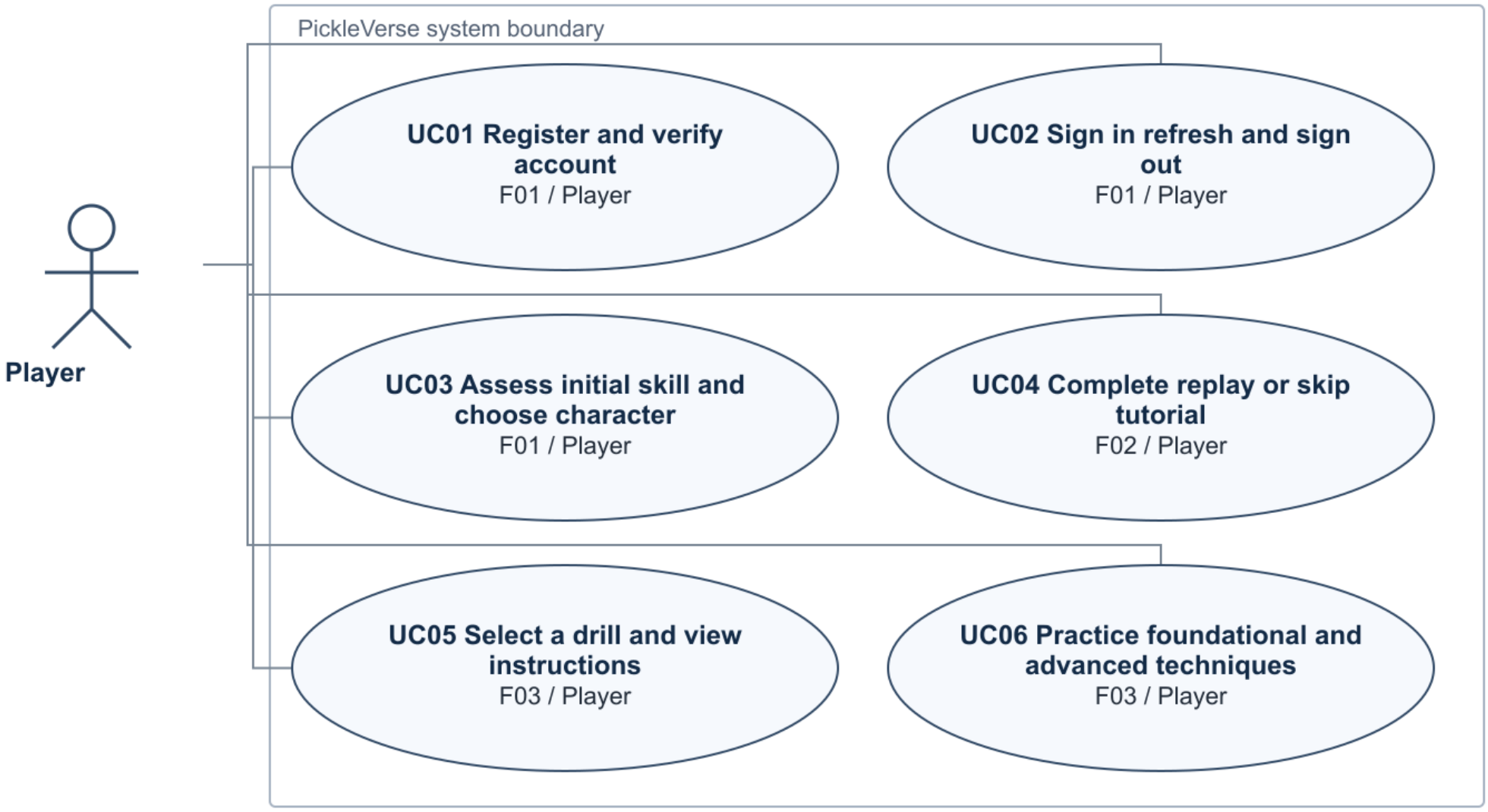
ARCH02 Proposal target with outstanding components

Target only. Dedicated simulation, ML evaluation, offline sync and operational guarantees are not claimed as completed.



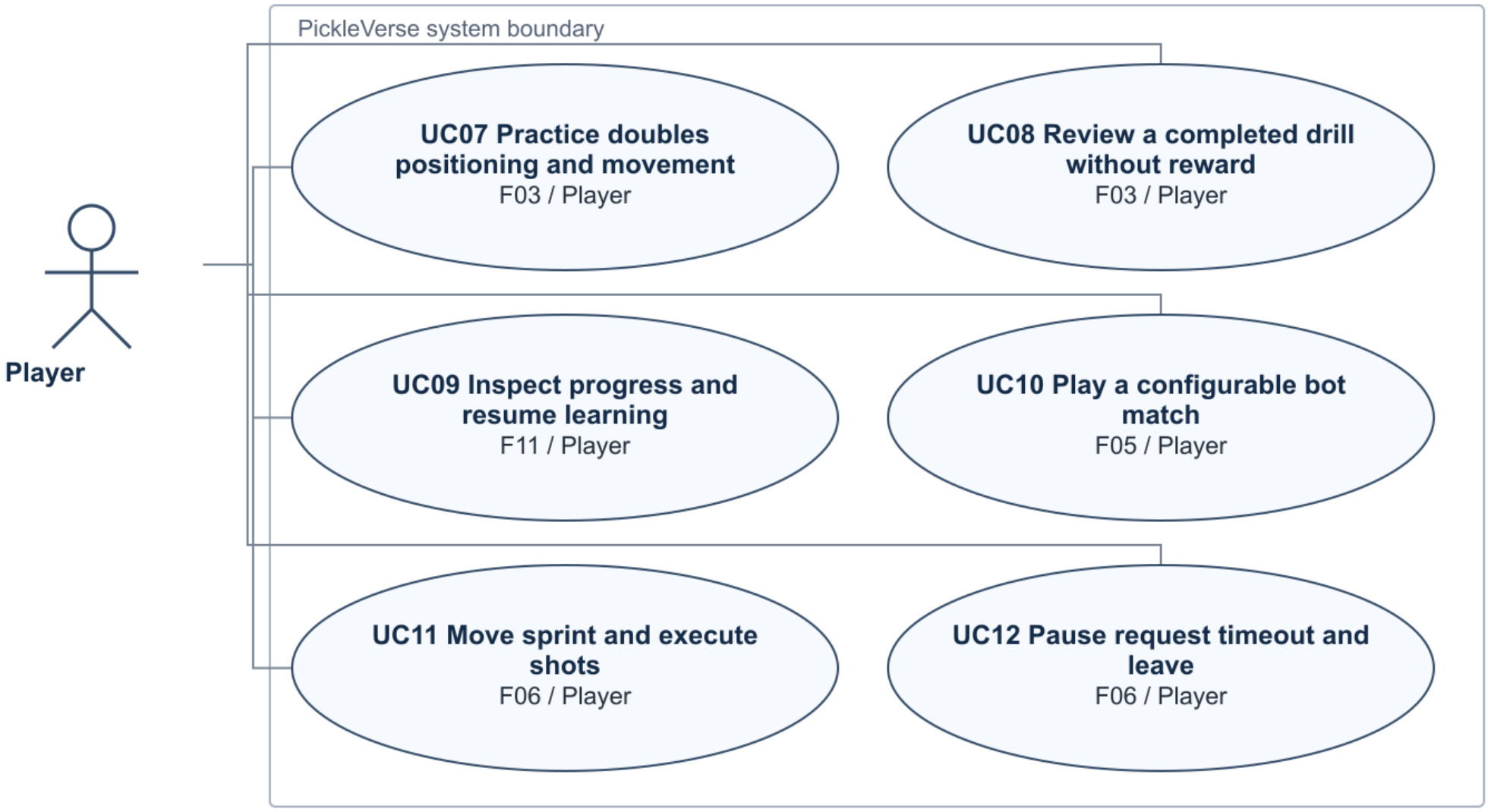
UC-D01 User goals UC01 to UC06

Goals are traceable to the SRS detailed specifications. Target/partial requirements are not claimed to be implemented.



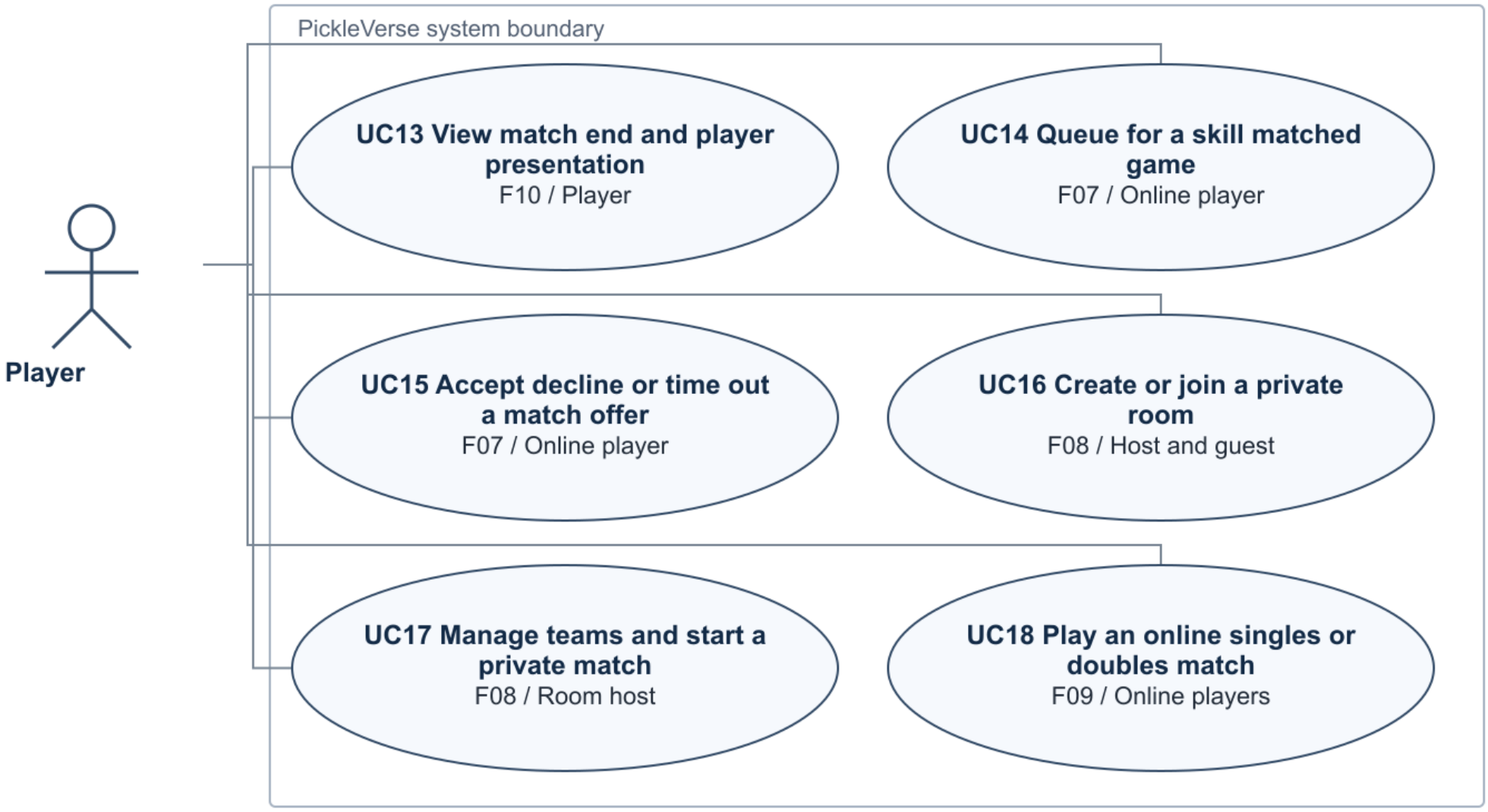
UC-D02 User goals UC07 to UC12

Goals are traceable to the SRS detailed specifications. Target/partial requirements are not claimed to be implemented.



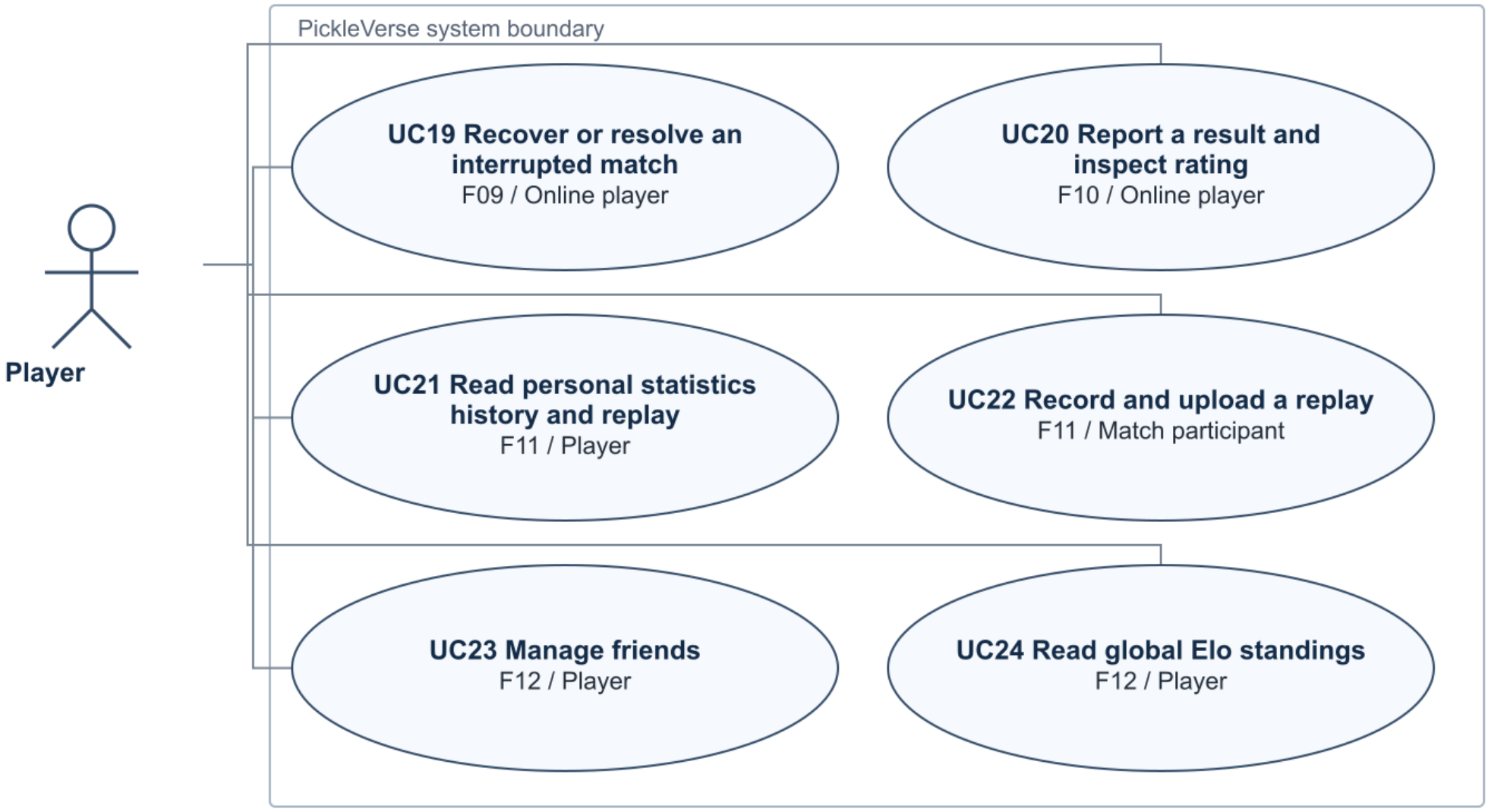
UC-D03 User goals UC13 to UC18

Goals are traceable to the SRS detailed specifications. Target/partial requirements are not claimed to be implemented.



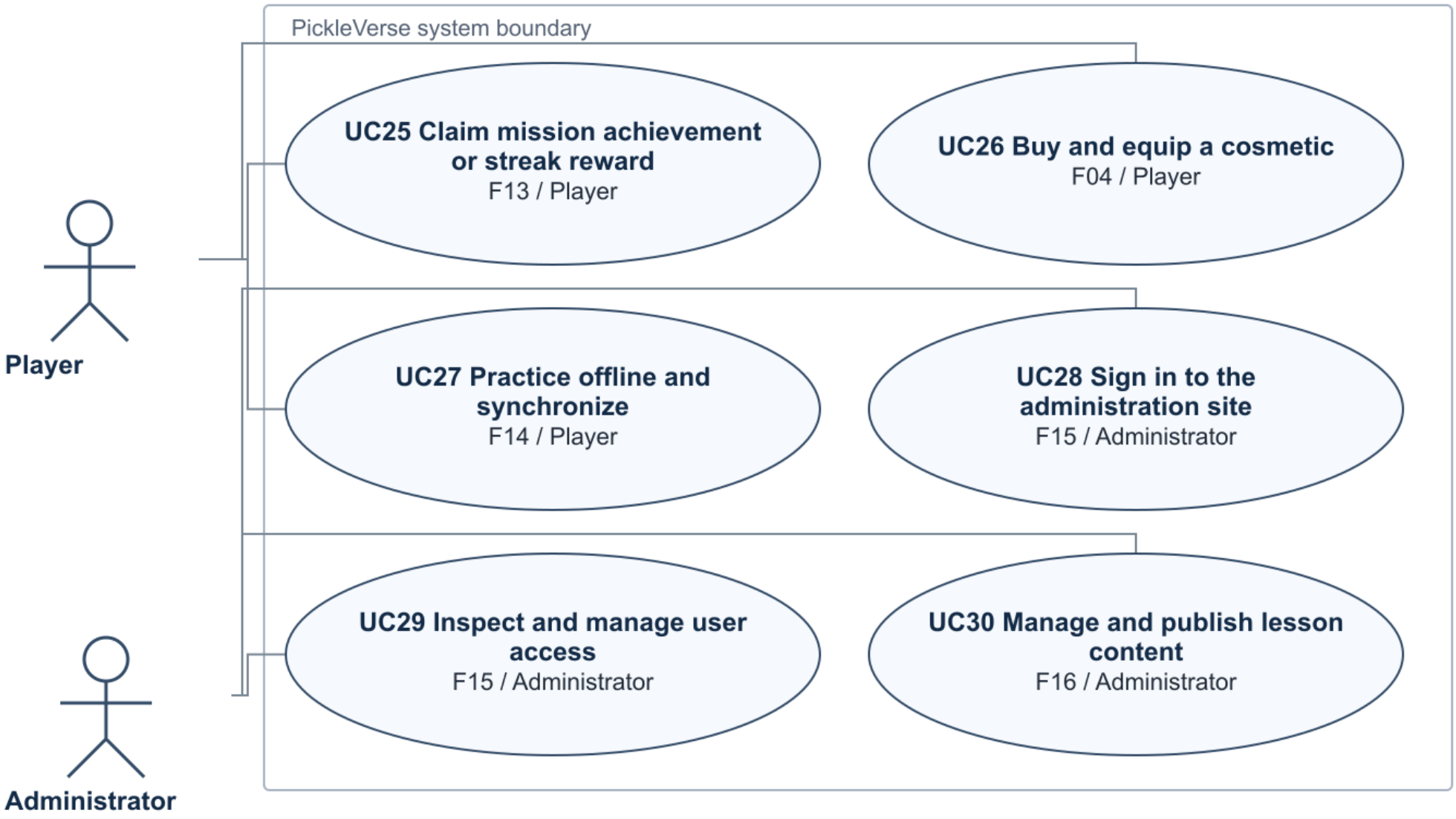
UC-D04 User goals UC19 to UC24

Goals are traceable to the SRS detailed specifications. Target/partial requirements are not claimed to be implemented.



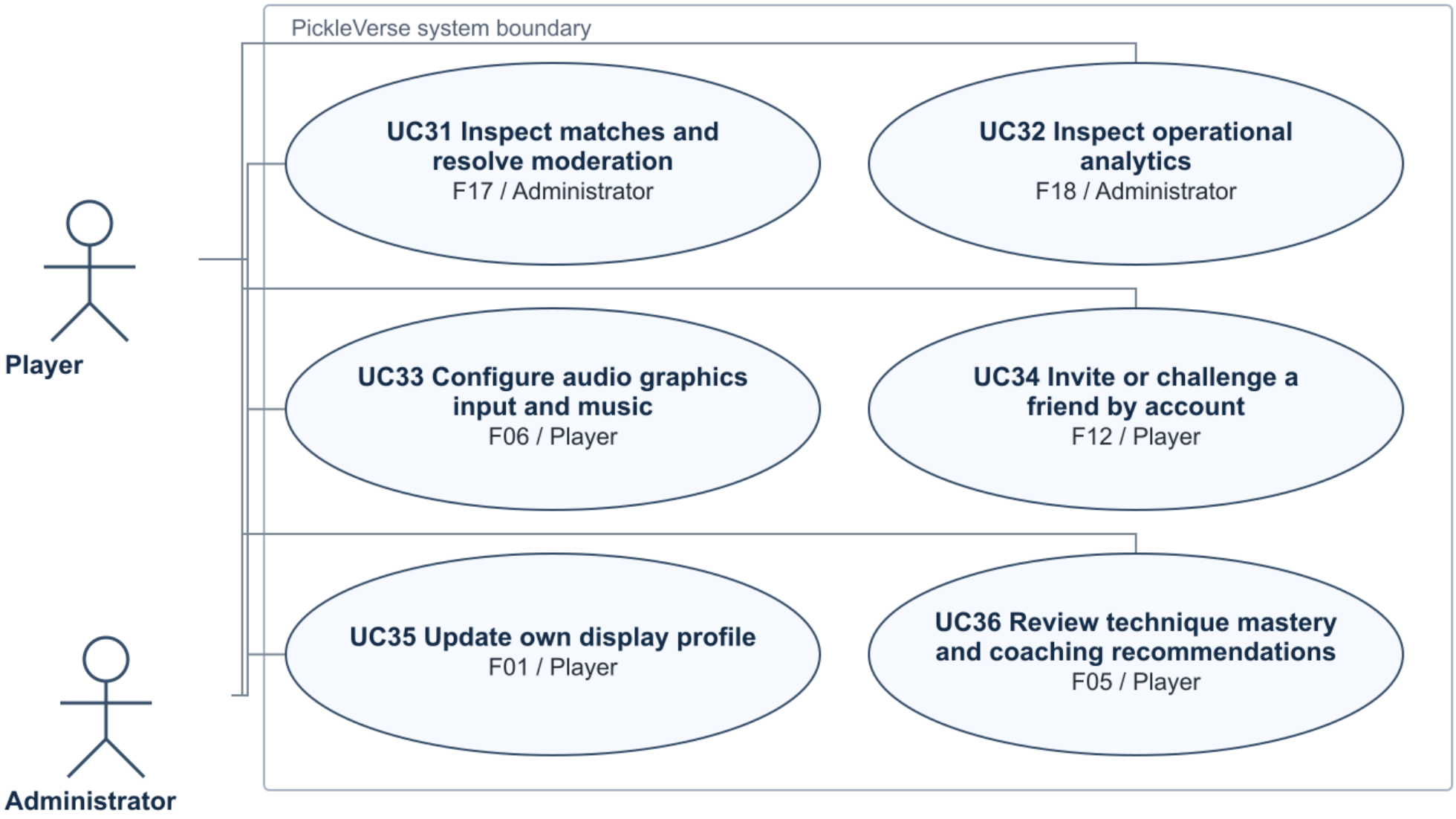
UC-D05 User goals UC25 to UC30

Goals are traceable to the SRS detailed specifications. Target/partial requirements are not claimed to be implemented.



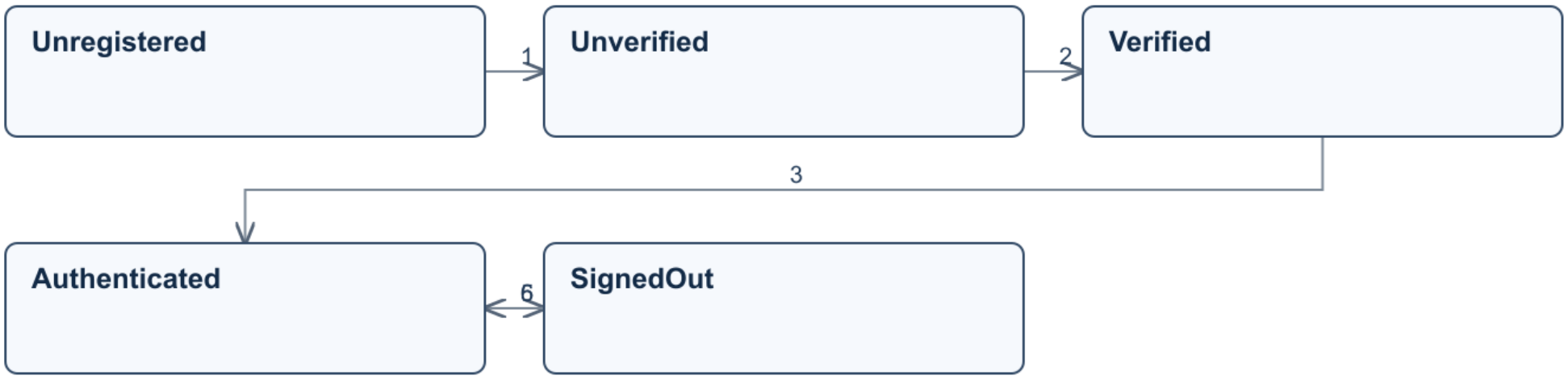
UC-D06 User goals UC31 to UC36

Goals are traceable to the SRS detailed specifications. Target/partial requirements are not claimed to be implemented.



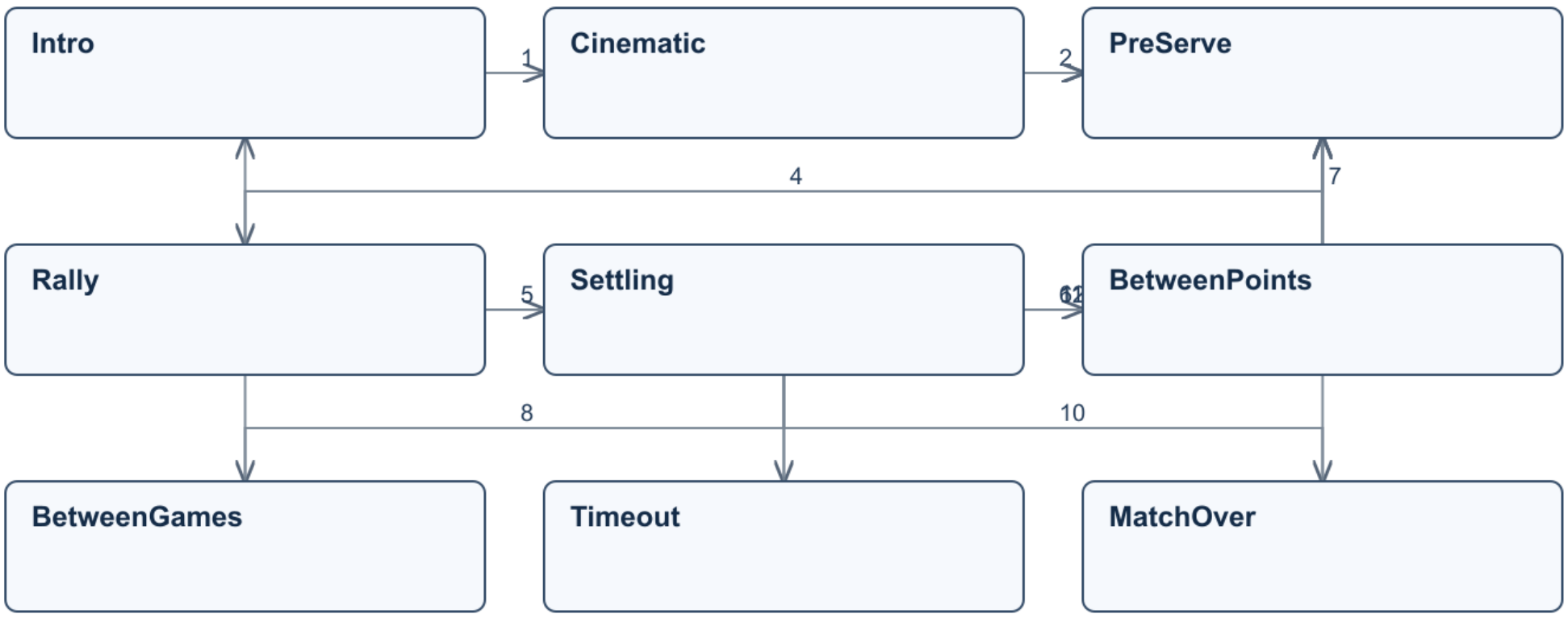
SM01 Account lifecycle

Derived lifecycle/guard model; do not equate labels with a database enum. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



SM02 Local match phase

Derived lifecycle/guard model; do not equate labels with a database enum. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



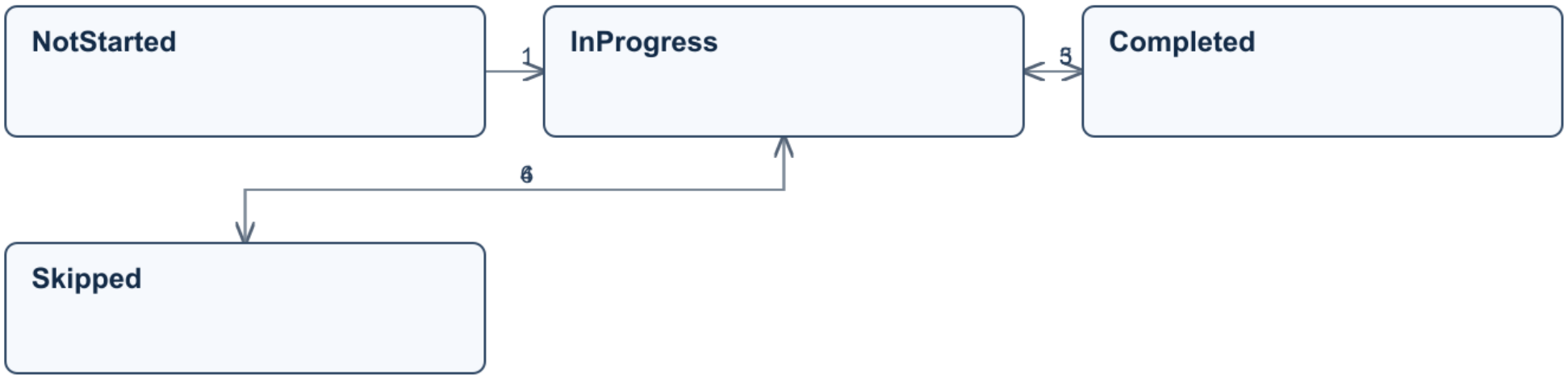
SM03 Training attempt

Derived lifecycle/guard model; do not equate labels with a database enum. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



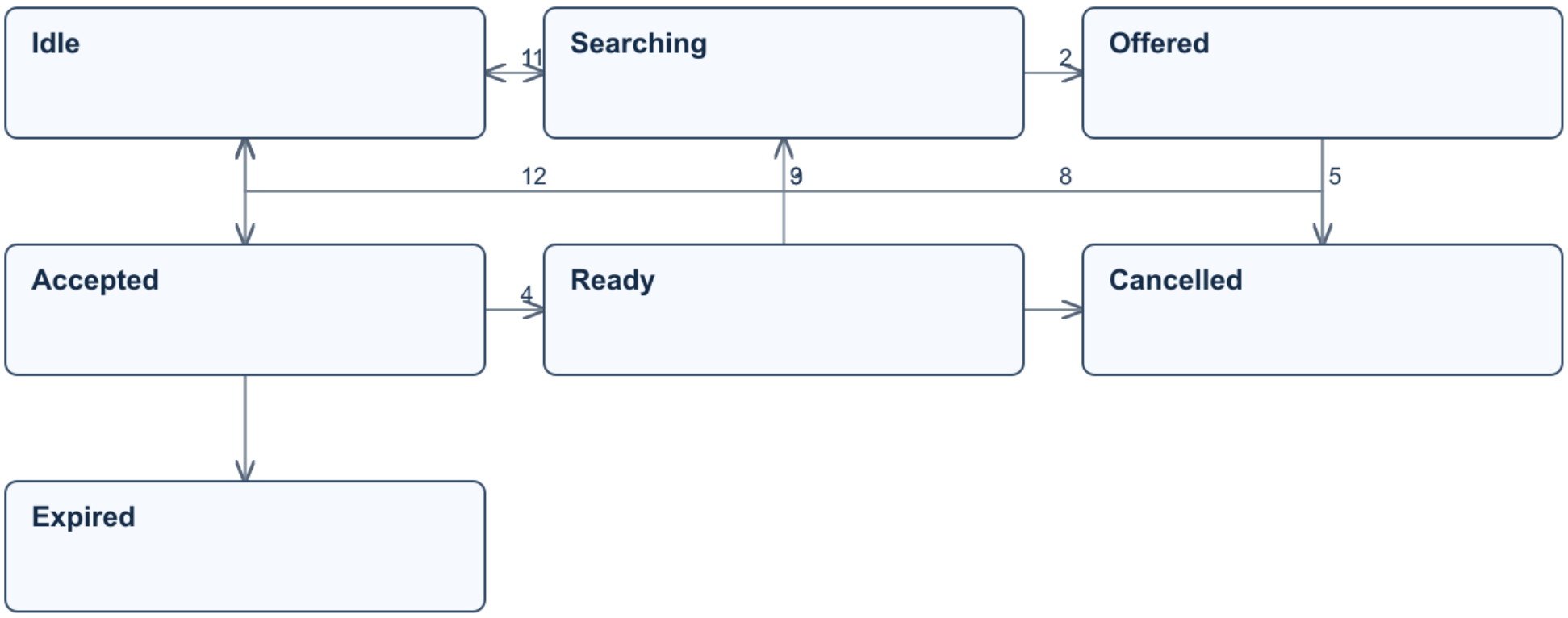
SM04 Tutorial progress

Derived lifecycle/guard model; do not equate labels with a database enum. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



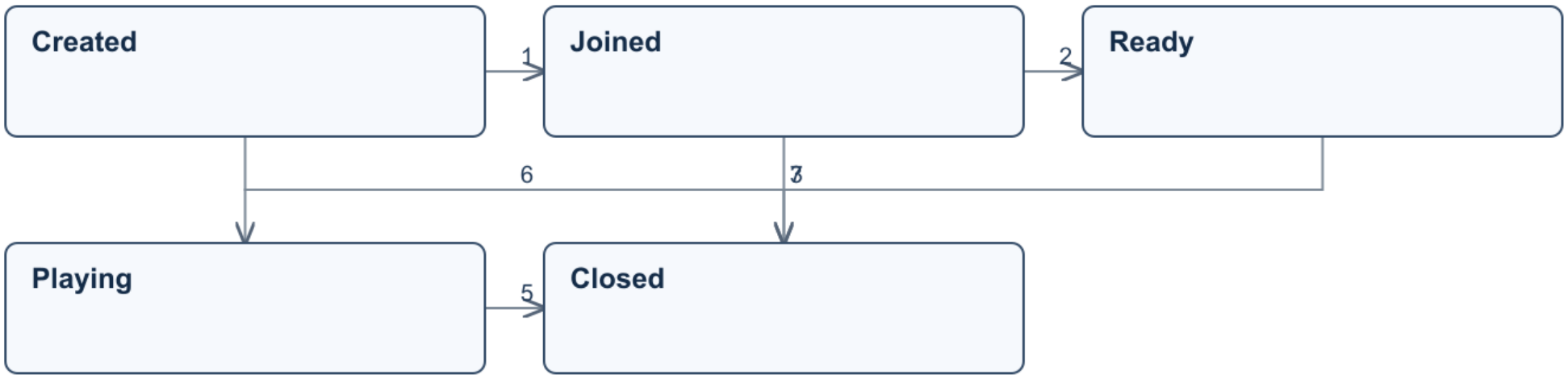
SM05 Skill queue and acceptance

Derived lifecycle/guard model; do not equate labels with a database enum. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



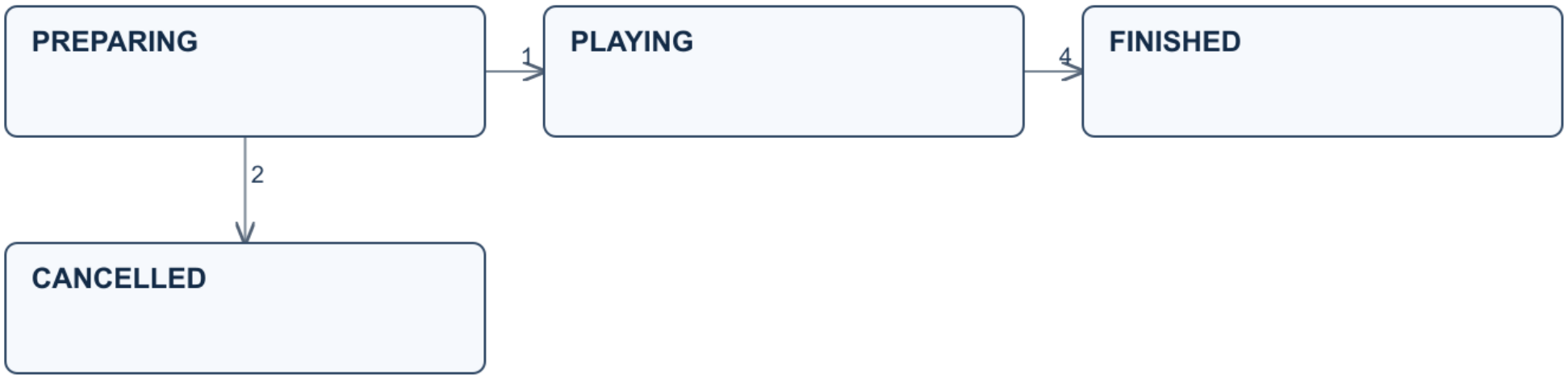
SM06 Private room

Derived lifecycle/guard model; do not equate labels with a database enum. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



SM07 Online canonical match

Derived lifecycle/guard model; do not equate labels with a database enum. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



SM08 Reward and command ledger

Derived lifecycle/guard model; do not equate labels with a database enum. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



SM09 Friendship

Derived lifecycle/guard model; do not equate labels with a database enum. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



SM10 Replay upload

Derived lifecycle/guard model; do not equate labels with a database enum. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



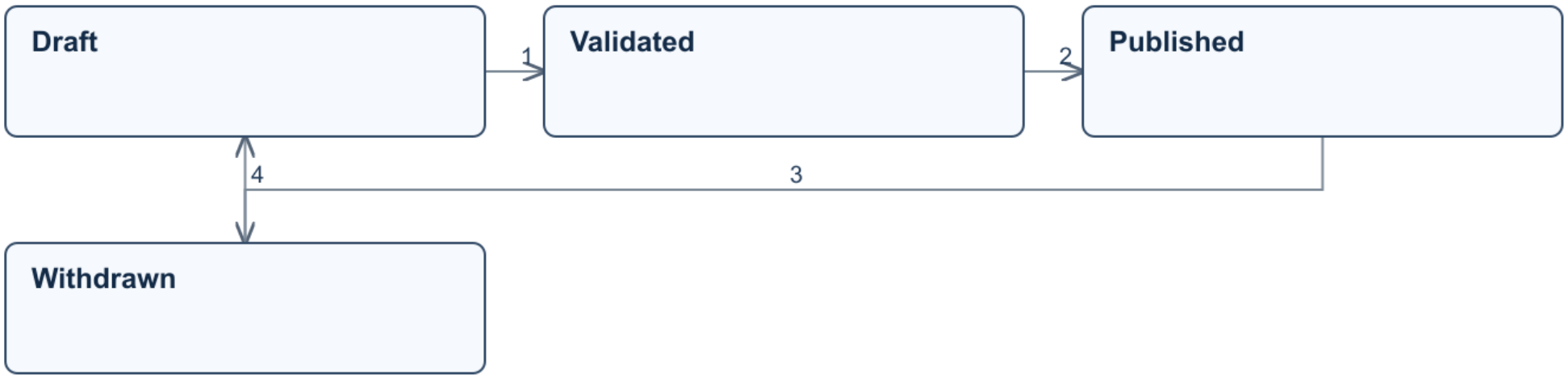
SM11 Administrative sanction

Target lifecycle, incomplete implementation; declared SQL states retained where applicable. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



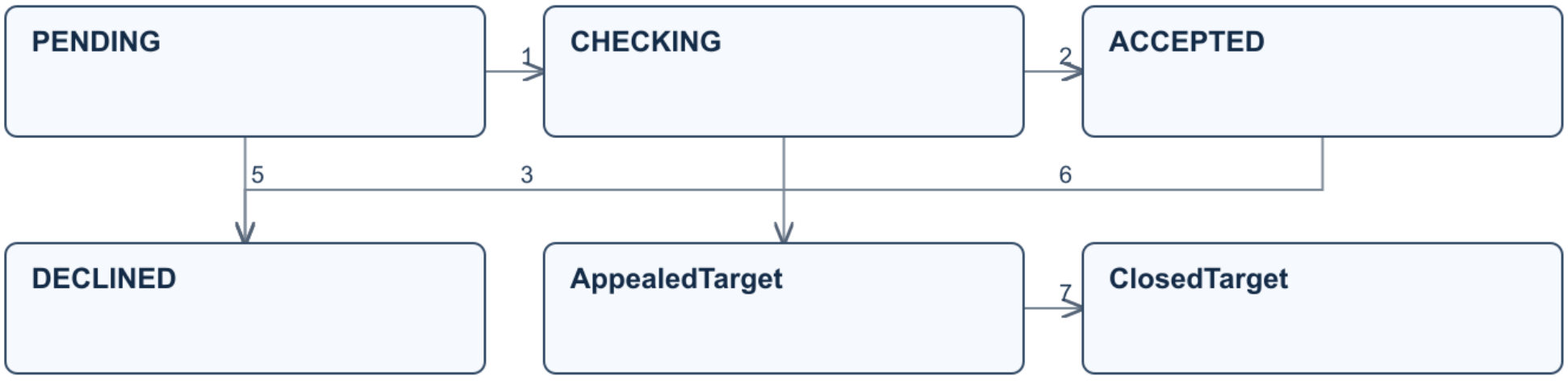
SM12 Lesson publication target

Target lifecycle, incomplete implementation; declared SQL states retained where applicable. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



SM13 Moderation and appeal target

Target lifecycle, incomplete implementation; declared SQL states retained where applicable. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



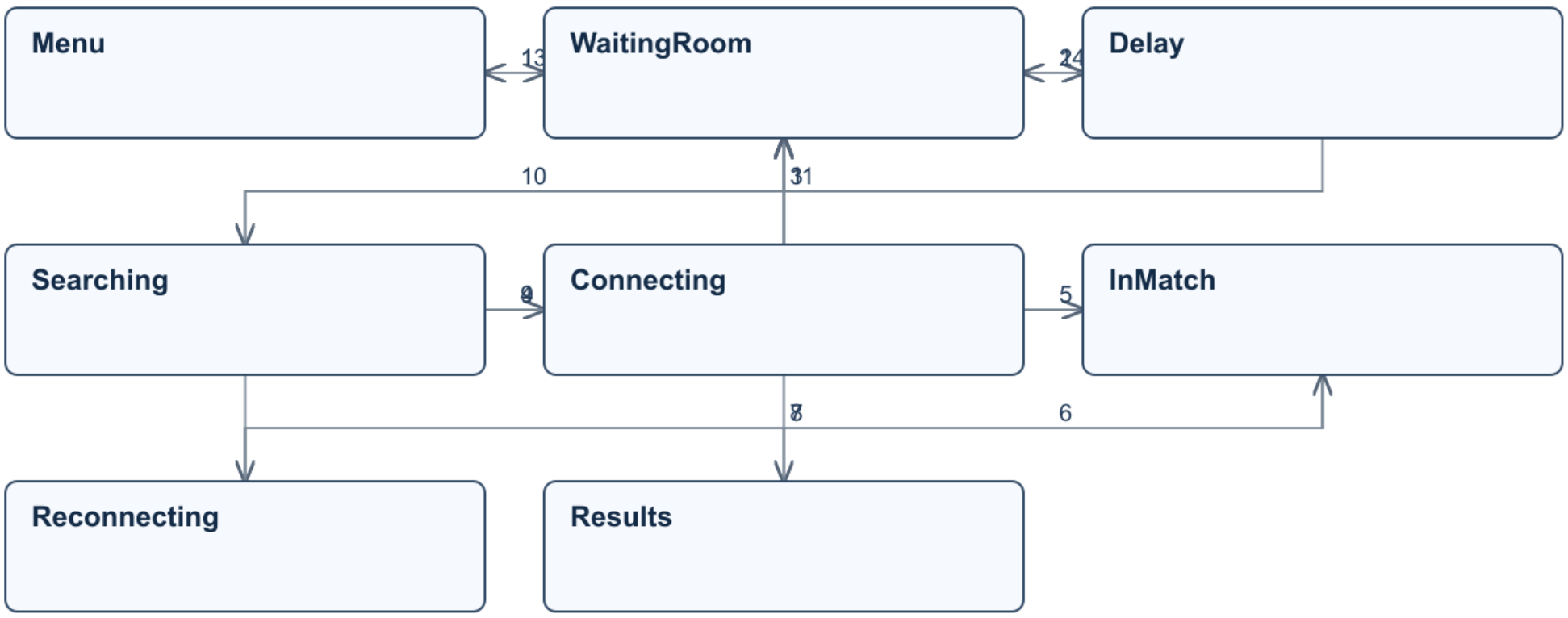
SM14 Training controller phase

Actual controller enum; transitions are selected main control paths. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



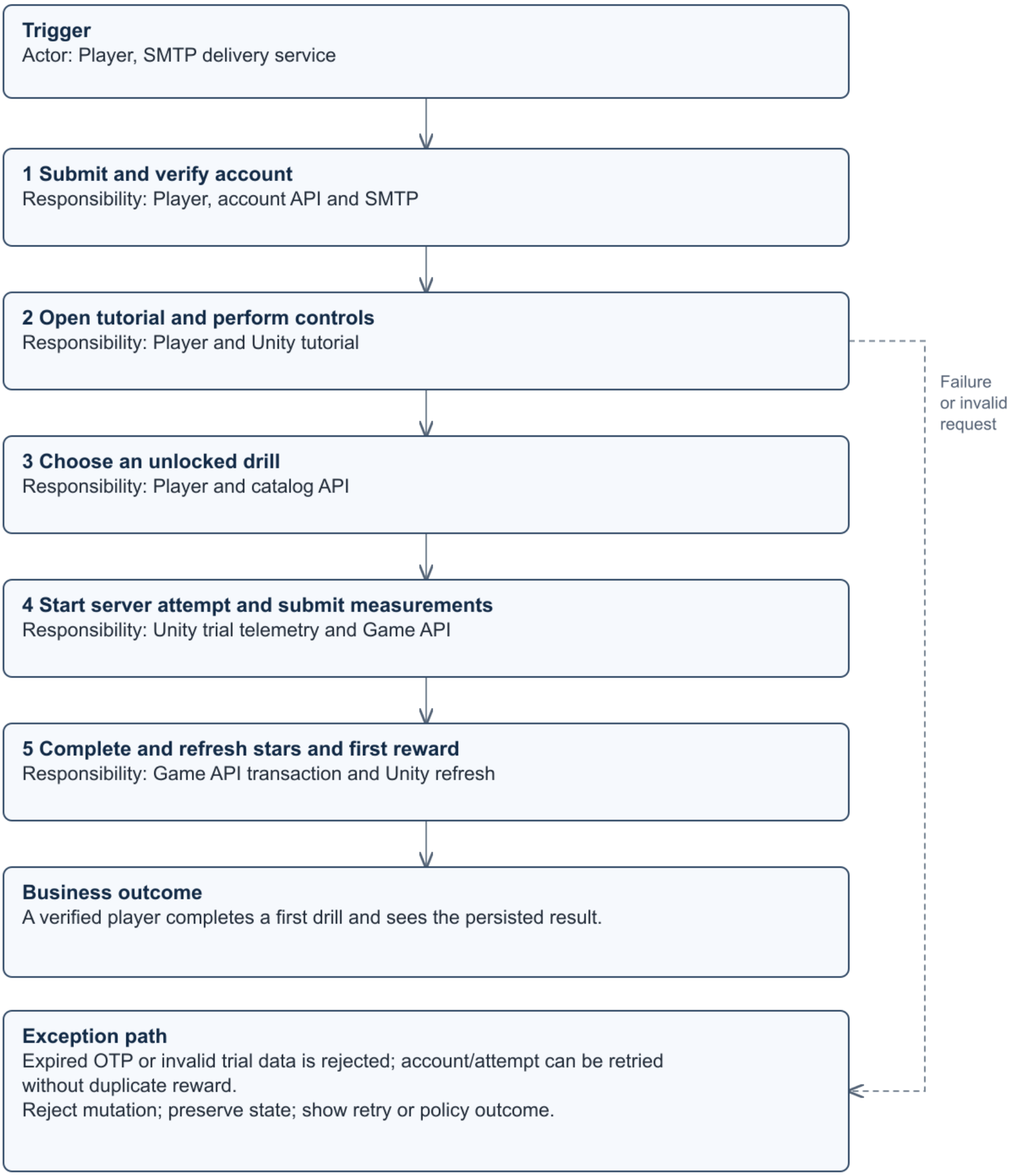
SM15 Multiplayer client flow

Actual FlowState enum; ready acceptance is a separate canonical match state. Numbered arrows map to the transition register in the SDD; rejected/self transitions are listed there.



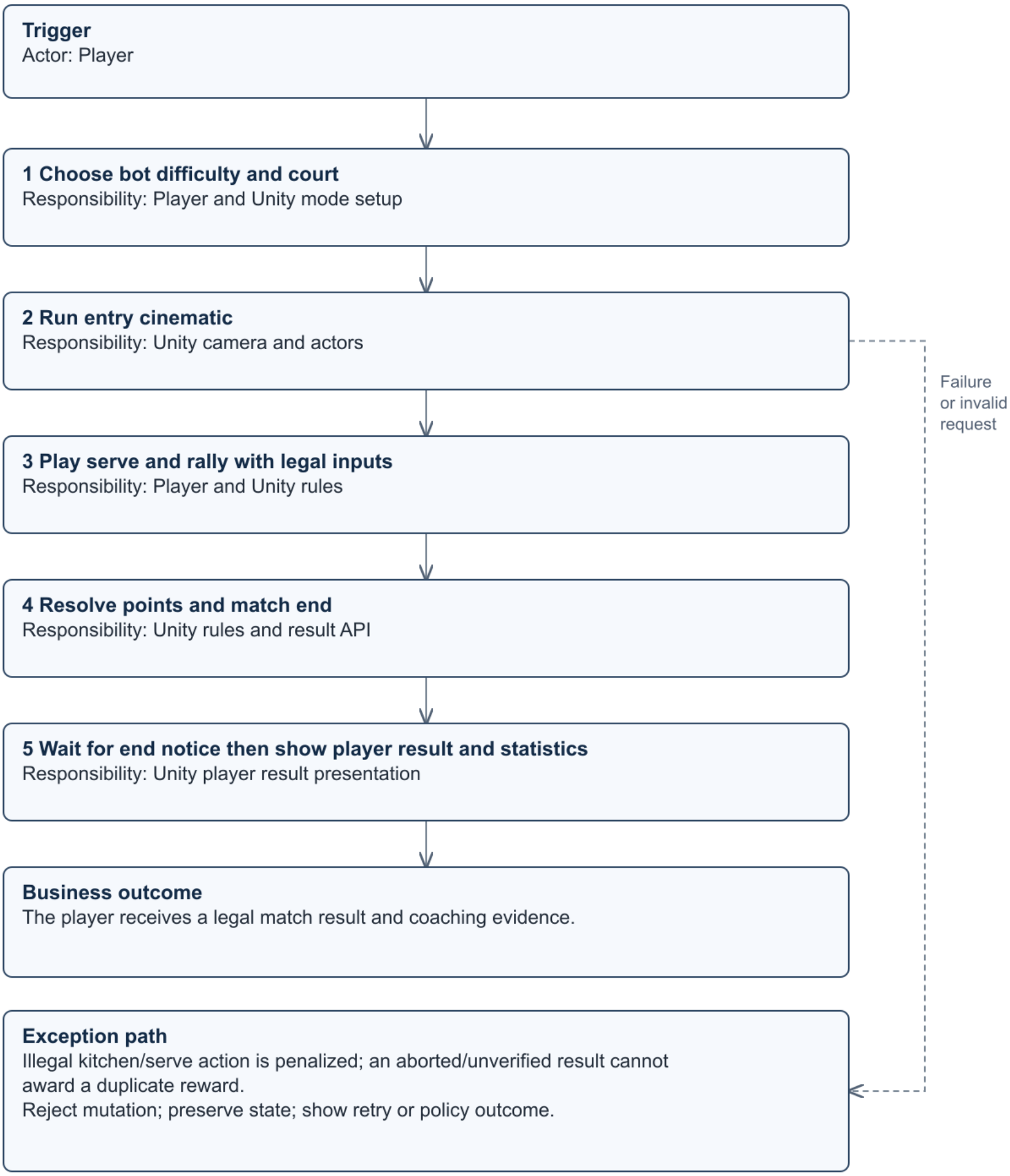
ACT-WF01 WF01 Register learn and complete a first drill

Owner slot M1 is proposed, not a named commitment. Actors are identified in the trigger; API/service responsibilities are internal.



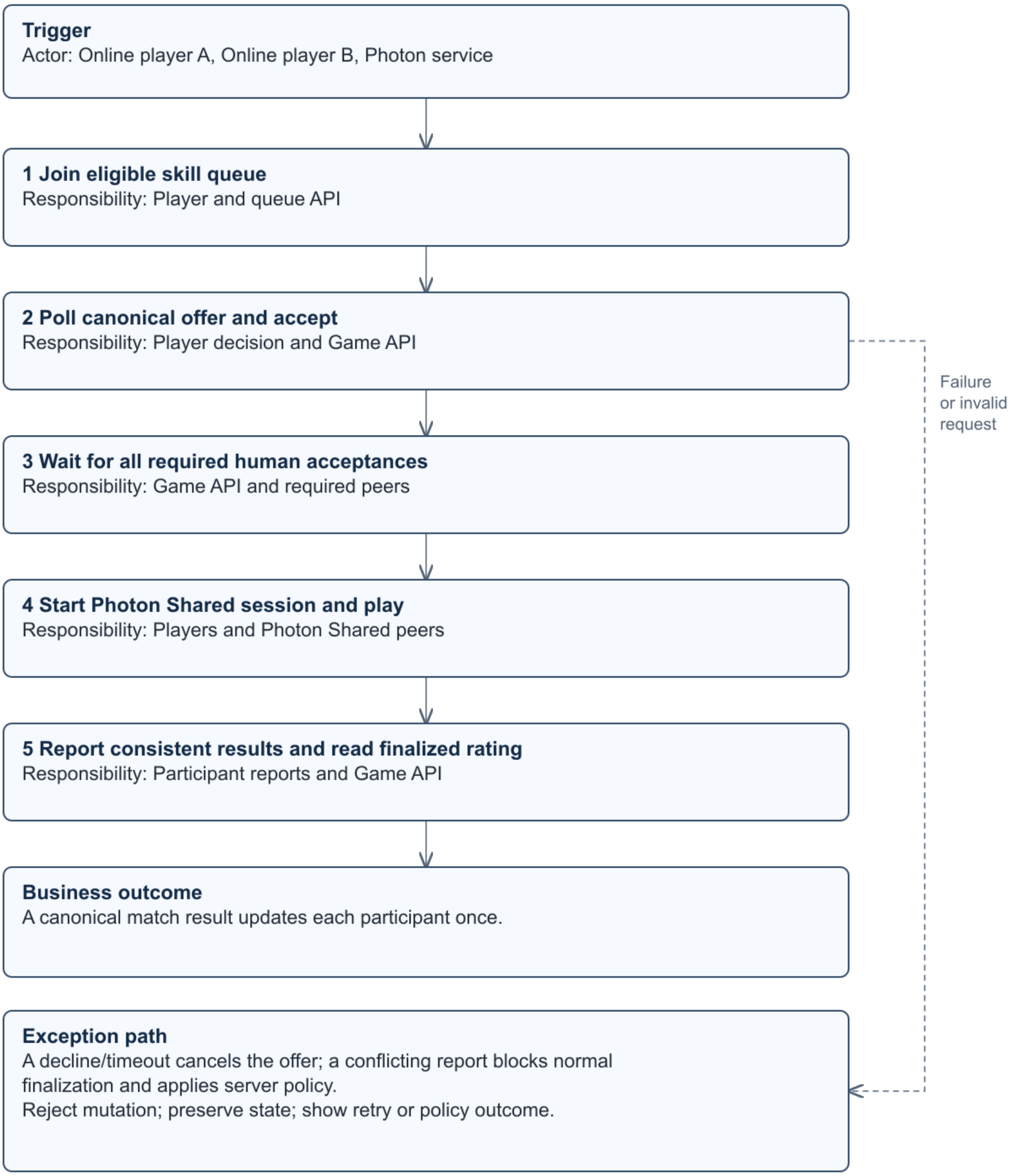
ACT-WF02 WF02 Practice against an adaptive bot

Owner slot M2 is proposed, not a named commitment. Actors are identified in the trigger; API/service responsibilities are internal.



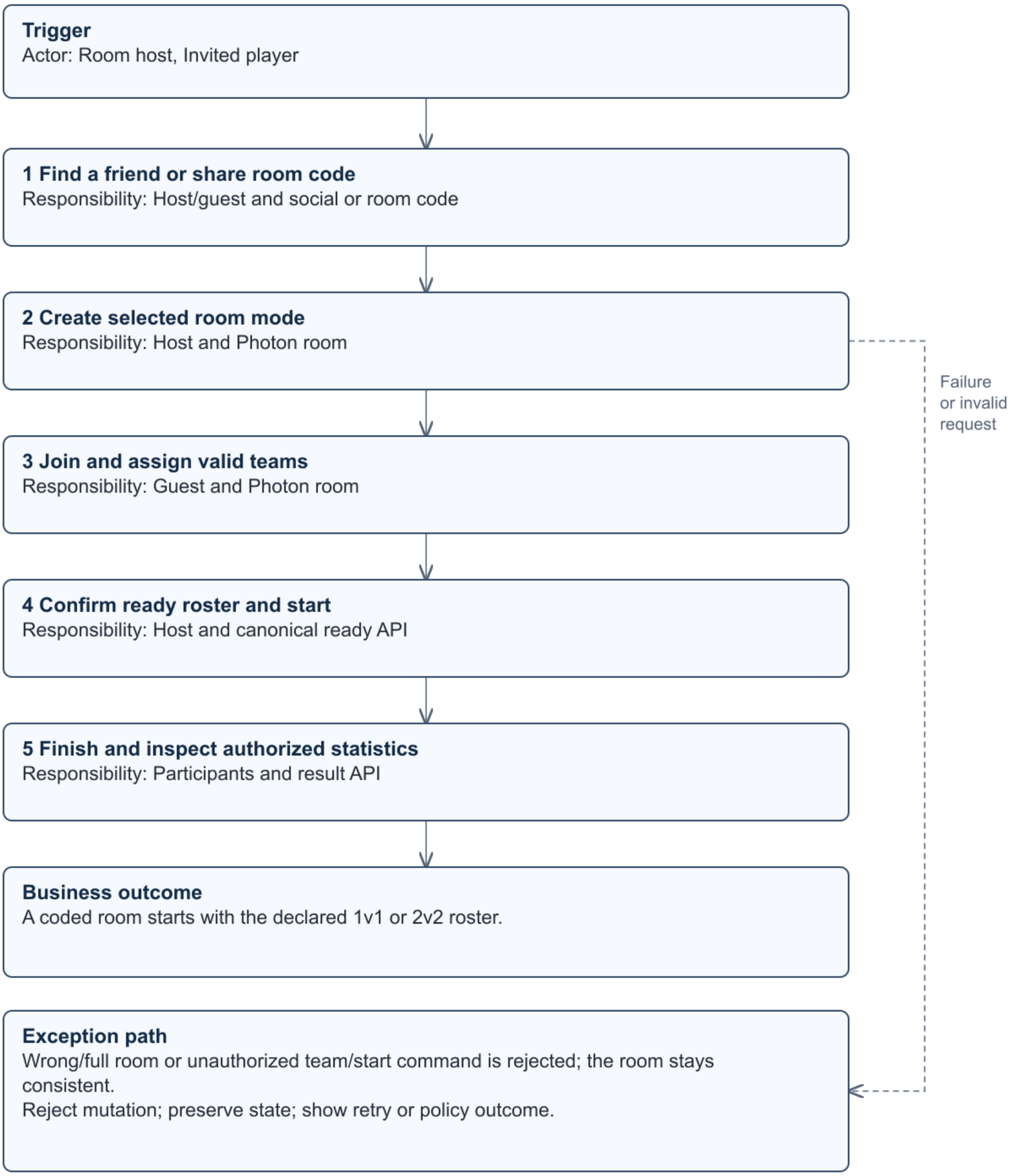
ACT-WF03 WF03 Find accept and finish an online match

Owner slot M3 is proposed, not a named commitment. Actors are identified in the trigger; API/service responsibilities are internal.



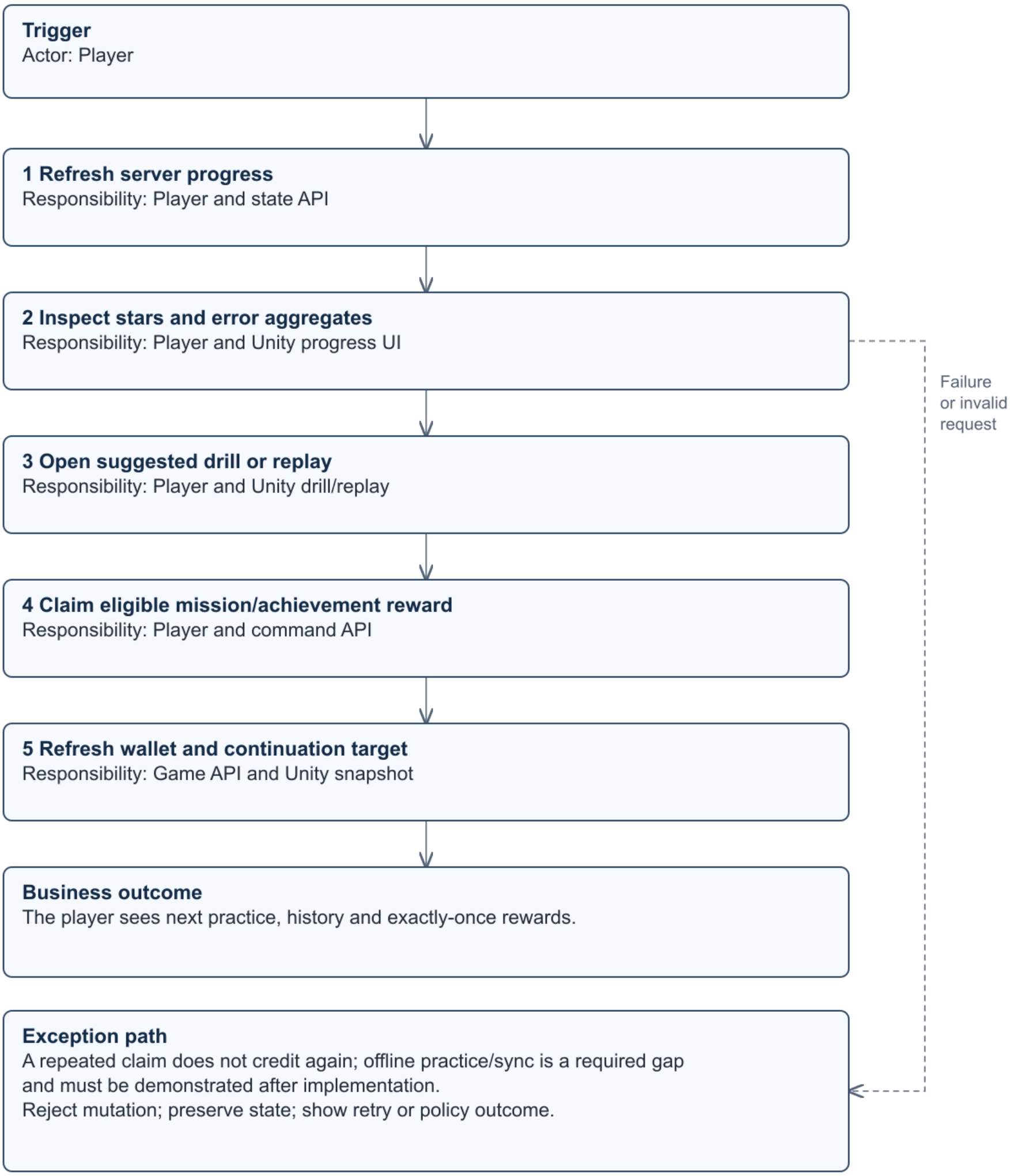
ACT-WF04 WF04 Create a private team match

Owner slot M3 is proposed, not a named commitment. Actors are identified in the trigger; API/service responsibilities are internal.



ACT-WF05 WF05 Review progress and claim rewards

Owner slot M1 is proposed, not a named commitment. Actors are identified in the trigger; API/service responsibilities are internal.



ACT-WF06 WF06 Administer users content and operations

Owner slot M4 is proposed, not a named commitment. Actors are identified in the trigger; API/service responsibilities are internal.

